Date: 12/11/2026

DSA03 – ASSIGNMENT 1

MORPHOLOGY-AWARE TEXT NORMALISATION AND SEARCH

Course Code: DSA03

Course Title: Natural Language Processing

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Assignment title: Build a morphology-aware text normalisation and search layer for a hospital record system using regular expressions, finite-state automata and morphological parsing.

Answer map and evidence conventions

This report answers Tasks 1–12 in sequence. It defines a finite-state baseline for clean typed English, specifies a controlled synthetic evaluation protocol, then revises the design for noisy OCR and Romanised code-mixing. Tables labelled "design" are specifications. Tables labelled "synthetic demonstration" are reproducible test fixtures. No number below is represented as a result from real hospital records.

- Corpus target: 500 synthetic records × 50 content tokens = 25,000 tokens, balanced over discharge summaries, prescriptions, operation notes and nursing observations. Each generated record has a stable record-type and concept label.
- Evaluation target: 40 query strings and query-to-record gold labels generated from known synthetic concept IDs. These are proxy judgements, not clinician judgements; a clinician review is required before deployment.
- Annotation target: 200 constructed surface forms with lemma/stem, part of speech and features, curated for testing regular and irregular forms.
- Effort values are engineering estimates, not time sheets; the suggested audit categories are included for later replacement with actual logged hours.

1. Task 1 – Frame the Problem: Models, Algorithms and Levels of Analysis

3 marks | Guiding enquiry: Which linguistic levels and model families are appropriate?

1.1 Levels of linguistic analysis

Level	What it studies	Hospital-search relevance and example
Phonology	Sound patterns and pronunciation.	Not required for clean typed English search. It becomes indirectly relevant to speech dictation and OCR confusion only if sound-based input is added; e.g., "ileum" versus "ilium" is not solved by spelling rules alone.
Morphology	Word structure and grammatical/derivational features.	Required: fracture, fractures, fractured and fracturing should be connected by inflectional analysis; nephrectomy and nephrectomies share number morphology, while renal is a semantic/concept relation rather than a safe affix rule.
Syntax	How words combine into phrases and clauses.	Light use is required to associate a value with a label and interpret "no fracture" versus "fracture present." Full parsing is outside the first retrieval layer.
Semantics	Meaning and relations among words and propositions.	Required for clinical synonym/concept expansion and negation-sensitive ranking; "renal" and "kidney" are related without being orthographic variants.
Pragmatics	Meaning in context, intent and shared knowledge.	Usually not needed for first-pass candidate retrieval, but needed for resolving shorthand such as "continue" in a plan versus a historical description.
Discourse	Links across sentences and document structure.	Useful for scope and temporal status ("history of…", "ruled out"), but not required to validate an isolated identifier or normalize a date.

The baseline therefore uses morphology and entity recognition, with only shallow syntax/semantics for section cues, negation and controlled synonym expansion. It does not claim that stemming understands clinical meaning. A search result is a candidate for review; the normalizer must preserve the original text and never overwrite source records.

1.2 Model families and task fit

Family	Best fit here	Why the alternatives are poorer
State machines	Validate regular structured strings: MRN, dates, phone numbers, vital-sign formats.	A probabilistic model is unnecessary for a fixed grammar and may accept impossible forms; a general rule system can work but gives less direct, testable language recognition.
Formal rule systems	Token normalization, inflectional spelling alternations, and finite-state lexicon/rule composition.	A plain state machine without lexical labels cannot map "fractures" to lemma + features. A statistical model needs labelled examples and may be harder to audit for deterministic spelling rules.
Probabilistic models	OCR correction ranking, ambiguous abbreviation resolution and noisy mixed-language token identification.	A fixed automaton cannot rank competing repairs; handwritten/OOV errors violate exact-match assumptions. A hand-written rule system grows brittle as noise and variation expand.

Ambiguity is central because the same observed string can support distinct analyses, and the intended interpretation depends on surrounding evidence. Lexical ambiguity: "lead" can be a verb or a metal-related noun in a clinical context. Morphological ambiguity: "found" can be a past-tense verb or a lexical adjective. Segmentation ambiguity: "12lead" may be "12-lead" or a malformed sequence.

Abbreviation ambiguity: “MS” can mean multiple sclerosis, morphine sulfate, or a department label. These are not all solved by a stemmer.

1.3 Search costs and pipeline

Automaton search follows transitions over a finite state set: for a deterministic automaton it is O(n) time and O(1) working memory per input string, aside from stored transition tables. NFA simulation is O(n|Q|) in a simple set-based implementation. Backtracking regular-expression engines can consume exponential time on pathological nested quantifiers; use anchored patterns, bounded quantifiers and non-backtracking/linear engines for untrusted text. A trie/FST lexicon consumes memory proportional to the encoded lexicon and offers fast lookups. Probabilistic ranking consumes model memory and inference time, plus annotated examples and expert review effort.

RAW RECORD / OCR TEXT
↓ UTF-8 decode, provenance + source offset retention [deterministic ingestion]
TOKENISE + boundary recovery [regular expression; maximal-munch]
↓
STRUCTURED ENTITY CANDIDATES [anchored regex + DFA validators]
↓
CASE / DATE / UNIT CANONICALISATION [finite rewrite rules; reversible map]
↓
LEXICON + TWO-LEVEL MORPHOLOGY [lexicon FST ∘ spelling rules]
↓
PORTER STEM AS LOW-PRIORITY FALLBACK [rule-based stemmer; O(n)]
↓
NEGATION / SECTION / LANGUAGE FLAGS [shallow rules; later weighted components]
↓
INDEX: original token + normalised lemma/features + entity fields
↓
QUERY ANALYSIS → candidate retrieval → ranking → clinician review

Input: four record types, initially clean typed English. Output: an auditable index retaining original text, offsets, entity values, lemma/features and confidence flags. OCR correction and token-level language identification are intentionally deferred to Task 12.

2. Task 2 – Basic Regular Expression Patterns

4 marks | Exact matching specifications and pattern inventory

2.1 Operators with minimal clinical examples

Operator	Meaning	Example / note
Concatenation	A followed by B.	[A-Z]{2}[0-9]{6}: two letters followed by six digits in an identifier.
Disjunction	A or B, grouped with parentheses.	(?:mg|mcg|g) accepts one supported mass unit.
Kleene star *	Zero or more repetitions.	[-]* allows zero or more spaces/hyphens only where explicitly permitted.
Plus +	One or more repetitions.	[0-9]+ matches an integer run.
Optional ?	Zero or one occurrence.	(?:\.[0-9]+)? permits a decimal fraction.
Character class	One character from a set.	[0-9] is one digit; [A-Za-z] is one ASCII letter.
Range	Contiguous characters.	[A-F] is a hexadecimal-letter range, not a semantic category.
Negated class	One character not in a set.	[^]+ is any non-linebreak run.
Anchors	Start/end of the tested string.	^…$ prevents a valid-looking substring from validating a whole field.
Counters	Bounded repetition.	[0-9]{4} requires exactly four year digits.

Python-compatible patterns use non-capturing groups (?:…), explicit ASCII digits [0-9], and fullmatch rather than relying on ^/$ behaviour around final newlines. The patterns below are candidate recognizers, not a medication dictionary or clinical validation rules.

2.2 Twelve structured-entity patterns

Patient identifier: (?i)\bMRN[-]?[A-Z0-9]{6,10}\b

- Matches MRN plus 6–10 alphanumeric characters, with optional hyphen/space. Must not accept a partial substring, 5-character suffix, or punctuation inside the identifier.

Admission date: \b(?:0?[1-9]|[12][0-9]|3[01])[-/.](?:0?[1-9]|1[0-2])[-/.](?:19|20)[0-9]{2}\b

- Day-first numeric dates; range checks still need calendar validation. Must not accept 32/13/2025 or mixed separators.

Discharge date: \b(?:0?[1-9]|[12][0-9]|3[01])[-/.](?:0?[1-9]|1[0-2])[-/.](?:19|20)[0-9]{2}\b

- Same lexical date grammar, but the field label/context determines discharge rather than admission. Must not infer event type from date shape alone.

Drug + strength: \b[A-Za-z][A-Za-z0-9-]{1,39}\s+[0-9]+(?:\.[0-9]+)?\s*(?:mg|mcg|g|ml|IU)\b

- Captures a drug-like name and strength/unit. Must not claim the first word is a verified drug; reject unsupported units or missing quantity.

Dosage frequency: \b(?:once|twice|thrice|[1-4]\s*times?)\s*(?:daily|a\s*day|per\s*day|q\.?\s*[0-9]+\s*h|BID|TID|QID|OD)\b

- A limited frequency vocabulary. Must not treat every uppercase token as a frequency; ambiguous OD requires context.

Lab value + unit: \b[A-Za-z][A-Za-z0-9-]{0,19}\s*[:=]?\s*[0-9]+(?:\.[0-9]+)?\s*(?:mg/dL|g/dL|mmol/L|10\^?[0-9]+/L|%)\b

- Label, numeric result and controlled unit. Must not infer the analyte or convert units without a validated mapping.

Blood pressure: \b(?:BP\s*)?[0-9]{2,3}\s*/\s*[0-9]{2,3}\s*(?:mmHg)?\b

- Systolic/diastolic pair. Must not accept zero, implausible limits, or a date-like pair without contextual validation.

Temperature: \b(?:Temp(?:erature)?\s*)?[0-9]{2}(?:\.[0-9])?\s*°?\s*(?:C|F)\b

- Temperature plus scale. Must not accept unitless numbers; plausible-range checks are a separate validator.

ICD code: \b[A-TV-Z][0-9][A-Z0-9](?:\.[A-Z0-9]{1,4})?\b

- ICD-like lexical shape. Must not imply that a code is valid for a particular ICD release; use a versioned code set.

Ward and bed: \b(?:Ward|W)\s*[A-Z0-9-]{1,8}\s*[,;/]+Bed\s*[A-Z0-9-]{1,8}\b

- Explicit ward and bed labels. Must not accept ward alone or a bed number detached from a ward context.

Contact number: \b(?:\+91[-]?)?[6-9][0-9]{9}\b

- Indian mobile-number-shaped string with optional +91. Must not accept fewer digits or claim active ownership.

Age: \b(?:Age\s*[:=]?\s*)?(?:0|[1-9][0-9]?|1[01][0-9]|120)\s*(?:years?|yrs?|y/o)\b

- Age 0–120 with unit. Must not accept 121 years or infer age from an unrelated number.

Symptom duration: \b(?:for\s+)?[0-9]+\s*(?:hours?|days?|weeks?|months?)\b

- Duration expression. Must not accept a bare number; surrounding syntax is required to associate it with a symptom.

  Date regex limitation: the grammar alone accepts 31/02/2025. The DFA recognizes shape; a Gregorian calendar check must reject impossible dates. The same distinction applies to clinically plausible values and versioned ICD codes.

2.3 Greedy, precedence and corpus errors

In “Give 500 mg at 08:00 and 250 mg at 20:00,” `.*mg` is greedy and can span from the first dose through the last `mg` if the pattern is not line-bounded. `.*?mg` stops at the earliest `mg` after the starting point. The quantifier changes the selected endpoint, not the clinical interpretation; prefer a bounded dosage pattern that consumes a number and unit and never crosses sentence boundaries.

Precedence is repetition, then concatenation, then alternation unless grouped. `^BP|HR$` means `(^BP) | (HR$)`, not “the whole string is BP or HR.” The intended pattern is `^(?:BP|HR)$`. A misplaced `|` can silently admit unrelated strings.

Illustrative error accounting protocol: a clinician-labelled gold span is compared with pattern spans using exact boundaries. TP = exact correct spans, FP = invalid/incorrect spans, FN = missed gold spans. Since no real records were supplied, we do not claim per-pattern corpus TP/FP/FN. The synthetic unit tests in Appendix B report expected accept/reject cases; before deployment, the same table should be populated on a held-out, de-identified, clinician-adjudicated sample. Error causes to code separately are OCR character substitution, separator variation, unlisted unit, abbreviation variation, boundary capture, invalid calendar date, label omission, and out-of-range value.

3. Task 3 – Regular Expressions for Tokenisation and Normalisation

4 marks | Token definition, maximal-munch examples and limits

3.1 Tokenizer specification

```
(?ix)
  (?:[A-Z]{1,5}\.){1,4}              # dotted abbreviations: e.g., i.v.
| [A-Za-z]+(?:[-'][A-Za-z0-9]+)*      # words, contractions, hyphenated forms
| [0-9]+(?:\.[0-9]+)?\s*(?:mg|mcg|g|ml|mmol/L|mmHg|°C|°F) # value+unit
| [0-9]+(?:\.[0-9]+)?                 # integer or decimal
| \+?[0-9][0-9() -]{6,}[0-9]          # phone-like sequence
| [^\s]                              # punctuation fallback
```

The specification is implemented with ordered alternatives and maximal-munch selection. In an actual scanner, the numeric+unit branch is tested before the bare number branch, the longest valid span wins, and ties use the branch order. Hyphens internal to compounds remain; sentence punctuation is a separate token. Dotted abbreviations are accepted only as a complete abbreviation, not as arbitrary runs of periods.

3.2 Maximal-munch demonstrations

Raw string	Whitespace split	Maximal-munch tokens	Why it matters
BP 120/80 mmHg; Temp 37.2°C	[BP, 120/80, mmHg;, Temp, 37.2°C]	[BP, 120/80, mmHg, ;, Temp, 37.2°C]	Punctuation is separated; a vitals value remains a useful unit-bearing token.
Take 0.5-mg tablet daily.	[Take, 0.5-mg, tablet, daily.]	[Take, 0.5, -, mg, tablet, daily, .]	Hyphen policy is explicit; a separate drug-strength parser may combine 0.5 mg where semantics support it.
Give i.v. 5 mL q.8h	[Give, i.v., 5, mL, q.8h]	[Give, i.v., 5, mL, q.8h]	Dotted abbreviation and interval remain intact rather than splitting at internal punctuation.
MRN-AB123456	[MRN-AB123456]	[MRN-AB123456]	Identifier grammar can recognise the whole entity and preserve its boundary.

3.3 Rewriting rules and information loss

Rule	Before → after	Reversibility and guard
Case fold	“FRACTURE” → “fracture”	Not reversible without storing the original; safe for case-insensitive retrieval.
Abbreviation expansion	“SOB” → “shortness of breath”	Many-to-one and context-sensitive; store original and expansion provenance; never expand “SOB” blindly.
Date canonicalisation	“12/11/26” → “2026-11-12”	Ambiguous month/day interpretation is not invertible unless locale/source metadata resolves it.
Unit canonicalisation	“0.5 g” → “500 mg”	Numerically reversible if precision and unit are retained; rounding can destroy precision.
Hyphen/line-wrap repair	“nephrec- tomy” → “nephrectomy”	May incorrectly merge intentional hyphens; preserve source offsets and confidence.

Thirty-abbreviation starter map, with expansion reserved for contexts where the abbreviation is unambiguous: BP=blood pressure; HR=heart rate; RR=respiratory rate; T=temp/temperature; SpO2=peripheral oxygen saturation; WBC=white blood cell count; Hb=haemoglobin; Hct=haematocrit; CXR=chest radiograph; ECG=electrocardiogram; CT=computed tomography; MRI=magnetic resonance imaging; US=ultrasound; IV=intravenous; IM=intramuscular; PO=oral; PRN=as needed; BID=twice daily; TID=three times daily; QID=four times daily; OD=once daily or right eye (ambiguous); NPO=nothing by mouth; Dx=diagnosis; Hx=history; Rx=prescription/treatment

(context-dependent); c/o=complains of; s/p=status post; ROM=range of motion; UTI=urinary tract infection; COPD=chronic obstructive pulmonary disease; CKD=chronic kidney disease; CAD=coronary artery disease. Maintain expansion alternatives rather than one unconditional string replacement.

Synthetic complete-record example: “DISCHARGE SUMMARY: MRN-AB123456 admitted 12/11/2026 with fractured femur. BP 120/80 mmHg; Temp 37.2 C. Given paracetamol 500 mg BID. No SOB. Discharged 15/11/2026.” → “discharge summary: mrn-ab123456 admitted 2026-11-12 with fractured femur. blood pressure 120/80 mmHg; temperature 37.2 °C. given paracetamol 500 mg twice daily. no shortness of breath. discharged 2026-11-15.” Dates are interpreted as day/month/year based on declared locale; original text and offsets remain stored.

A requirement outside regular-expression power is matching arbitrarily nested, balanced parentheses in an unrestricted note. The language of balanced parentheses is not regular: a finite automaton cannot store unbounded nesting depth. A bounded-depth regex is only an approximation. Syntax scope, semantic negation and the intended sense of an abbreviation also cannot be inferred from token shape alone.

4. Task 4 – Finite State Automata: Design, Non-determinism and Equivalence

5 marks | Four deterministic validators plus optional-part NFA

4.1 Formal definition and deterministic validators

A deterministic finite automaton is M = (Q, Σ, δ, q_0, F), where Q is a finite state set, Σ is the input alphabet (often character classes), δ: Q×Σ→Q is a total transition function, q_0 is the start state and F⊆Q is the accepting-state set. The dead state is included to make δ total. The four diagrams below use class-labelled transitions; "other" means every character not in the listed class.

Validator / diagram sketch	Transition-table summary
MRN q0 --M--> q1 --R--> q2 --[-]?--> q3 --[A-Z0-9]×6..10--> qA; invalid → qD	q0 on M → q1; other inputs → other→qD. Accept only states marked A; calendar/value range checks are post-automaton.
MRN continued q1 --R--> q2; q2 --[-]--> q3 or [A-Z0-9]--> q4; q4..q10 accept only after total 6..10 chars	q3 on alnum → q4..q10; other inputs → other→qD. Accept only states marked A; calendar/value range checks are post-automaton.
BP pair q0 --digits(2..3)--> qS --/--> qD1 --digits(2..3)--> qA; spaces are permitted around slash	q0 on digit → q1..q3; other inputs → other→qD. Accept only states marked A; calendar/value range checks are post-automaton.
Temperature q0 --label?--> q1 --2 digits--> q2 --(. digit)?--> q3 --degree?--> q4 --C|F--> qA	q0 on label/number → q1..q4; other inputs → other→qD. Accept only states marked A; calendar/value range checks are post-automaton.
Day-first date q0 --day digits--> qD --sep--> qM --month digits--> qY --same sep--> qYR --4 year digits--> qA	q0 on day digit → q1..q2; other inputs → other→qD. Accept only states marked A; calendar/value range checks are post-automaton.

For a production DFA table, each abbreviated state range expands into digit-count states; transitions on an allowed character advance one state, while disallowed characters go to qD. qD loops to itself. MRN's optional separator is represented by a branch in the NFA and subset construction below; the DFA counts the total identifier characters and permits at most one separator after the prefix. This construction is deterministic and finite, although a table grouped by character classes is more readable than 128 individual ASCII columns.

4.2 Optional-country-code telephone NFA

Language: optional "+91" and optional spaces/hyphens around the country code, followed by exactly ten digits; the first mobile digit is 6–9. NFA states are q0 start, q1 plus seen, q2 first country digit 9, q3 country-code 91, q4 separator, q5 first national digit, q6–q14 remaining digits, qF accept. ε-transitions branch from q0 to q5 (no country code) and to q1 (country code).

Input prefix	ε-closure/live states	Result
ε	{q0,q5}	Both local-number and country-code paths are available.
+	{q1}	Country-code path only.
+9	{q2}	Prefix "+9" is incomplete but still live.
+91	{q3,q4,q5}	Can consume an optional separator, or begin national number.
+91 9876543210	qF reached after 10 national digits	Accepted.
987654321	q5..q13, no qF	Rejected at end: only nine national digits.
+91-5876543210	dead/none after first national digit	Rejected: first mobile digit must be 6–9.

Subset construction turns each ε-closed NFA state set into one DFA state. For this NFA the reachable subset states are S0={q0,q5}, S1={q1}, S2={q2}, S3={q3,q4,q5}, S4={q6}, S5={q7}, …, S13={q14}, plus SD=∅. On +, S0→S1; on digit 6–9, S0→S4; on 9, S1→S2; on 1, S2→S3; permitted separator loops from S3; ten national digits advance S3/S4 to the final accepting subset. All other inputs go SD. Acceptance is any subset containing qF. The fully expanded implementation table is compactly specified by the digit-count index and the dead-state rule; the generated test suite enumerates all states.

Minimisation merges equivalent states only when they have identical future acceptance behaviour. The prefix states S1/S2/S3 cannot merge because their accepted continuations differ; similarly each national-digit-count state is distinguishable by the exact number of digits remaining. The dead state

remains distinct. In this design the count DFA stays at 16 reachable states including dead (with a separate optional separator state, 17); minimisation cannot reduce the digit-count chain, but may merge equivalent separator/epsilon-branch states if their right languages match. Report the exact count from the implementation if the precise alphabet/state encoding changes.

4.3 Thompson construction, reverse regex and pumping lemma

Thompson's construction creates a two-state fragment for each literal/class, connects concatenation fragments with ε, and creates split/join states for alternation and Kleene star. For `BP[ ]*[0-9]{2,3}/[ ]*[0-9]{2,3}`, compile literals B, P, optional space-loop, two required digit fragments, one optional third digit fragment, slash, optional spaces, and the second 2–3 digit group. Concatenating fragments yields an NFA with ε branches for optional spaces and the third digit. Induction over regex structure establishes that each fragment accepts exactly the language of its subexpression; hence the concatenated NFA accepts exactly the pattern language.

A regular expression can be derived from an automaton by state elimination. Add new start and final states; when eliminating k, update each remaining edge $i \to j$ with $R_{ij} := R_{ij} \mid R_{ik}(R_{kk})^*R_{kj}$. Eliminating the intermediate states of the MRN counter yields the regular-language description `MRN[- ]?[A-Z0-9]{6,10}`. The automaton-to-regex and regex-to-automaton constructions preserve accepted strings in both directions by construction.

Pumping lemma: if L is regular, there is p such that every $s \in L$ with $|s| \geq p$ can be split $s=xyz$ with $|xy| \leq p$, $|y|>0$, and $xy^iz \in L$ for every $i \geq 0$. Let $L=\{\text{“(”}^n\text{“)”}^n : n \geq 0\}$ with spaces omitted in the formal strings, i.e. $L=\{(^n)^n\}$. Suppose regular with pumping length p and take $s=(^n)^n$. Since $|xy| \leq p$, y contains only opening parentheses; $y=(^k)$, $k>0$. Pump $i=0$ to remove k openings while retaining p closings, so the result is unbalanced and not in L, contradiction. Therefore unbounded balanced nesting is not regular, matching Task 3's boundary.

5. Task 5 – Inflectional Morphology

3 marks | Paradigms, spelling rules, irregular forms and measurement

5.1 Terms and inflection/derivation distinction

A morpheme is the smallest conventional unit pairing form and meaning/function. A stem is the base to which an inflection attaches; an affix is a bound morpheme attached before or after a base. Inflection changes grammatical features without creating a new lexeme (fracture → fractures); a paradigm is the set of forms belonging to one lexeme across grammatical features. Derivation more often changes lexical meaning/category (infect → infection), is less uniformly productive, and commonly occurs closer to the root than outer inflection: infect + ion + s → infections, not infect + s + ion.

Paradigm	Illustration	Features
Noun number	fracture / fractures; kidney / kidneys	SG / PL
Verb tense/aspect	treat / treats / treated / treating	base, 3SG, past/participle, progressive
Agreement	the patient improves; patients improve	3SG present vs plural/base
Adjective comparison	severe / severer / severest (rare clinically); mild / milder / mildest	positive/comparative/superlative; often periphrastic for longer forms
Adverb comparison	rapidly / more rapidly / most rapidly	periphrastic; usually not an inflectional suffix
Possessive and contractions	patient's; isn't	possessive clitic and contractions are tokenisation-relevant; treat separately from derivation

5.2 Five clinical lexeme paradigms

Lemma/POS	Paradigm (selected forms)	Notes
fracture / N	fracture, fractures	Regular plural; search should equate number variants.
treat / V	treat, treats, treated, treating	Final t is not doubled in treated/treating.
admit / V	admit, admits, admitted, admitting	CVC doubling under stress/productivity conditions.
injury / N	injury, injuries	Consonant+y becomes ies; vowel+y takes s: days.
diagnosis / N	diagnosis, diagnoses	Sibilant/Greek plural; not reliably obtained by a universal -es rule.
severe / Adj	severe, severer/more severe, severest/most severe	Both synthetic and periphrastic comparison; style and syllable constraints matter.

5.3 Orthographic rules and over-generation

Rule (implementation sketch)	Example	Counterexample / guard
e-deletion: final silent e drops before vowel-initial suffix	treat + ing → treating	dye + ing → dyeing; protect lexical exceptions and e that represents a pronounced vowel.
CVC doubling before -ed/-ing for stressed final syllable	admit + ed → admitted; refer + ing → referring	visit → visiting, not visitting; do not double when final stress is absent or consonant is w/x/y.
y→i before suffix beginning with a vowel (with lexical constraints)	injury + s → injuries; study + ed → studied	play + ed → played; preserve y after vowel.
sibilant -es after s, x, z, ch, sh	dress + es → dresses; watch + es → watches	gas + s? Spelling depends on lemma; do not append -es to every word ending in a similar sound where the word is not a count noun.
f/fe plural alternation, lexically restricted	knife → knives	roof → roofs; must be listed or lexically marked.
irregular ablaut/suppletion	was/were; went; children	No productive concatenative rule derives these reliably; list lexical pairs.

Irregular clinical-relevant forms to list: be–am/is/are/was/were/been; have–has/had; do–does/did; go–went/gone; take–took/taken; give–gave/given; make–made; see–saw/seen; tooth–teeth; foot–feet;

diagnosis–diagnoses; analysis–analyses; child–children; man–men; woman–women; mouse–mice; datum–data (domain style varies); bacterium–bacteria; fungus–fungi; radius–radii; criterion–criteria. Rule-based endings cannot predict these without lexical class or exception entries.

The intended ratio is number of distinct surface types divided by number of lemma types over the same tokenized corpus. It must be computed from a documented corpus after tokenization. In a 200-form constructed annotation set, the denominator and numerator are known by construction, but that is not a corpus statistic. For an evaluation report, mark the share of missed query-relevant records attributable solely to inflection by comparing gold exact-match misses against lemma-normalized retrieval. The assigned 40-query synthetic set can estimate a controlled effect; real clinical rates require clinician judgement.

6. Task 6 – Derivational Morphology

4 marks | Clinical affixes, families, productivity and search policy

6.1 Affix inventory

Affix	Typical category effect	Core contribution	Clinical example
-itis	N→N	inflammation	appendicitis
-ectomy	N→N	surgical removal	nephrectomy
-otomy	N→N	incision	tracheotomy
-ostomy	N→N	opening/stoma creation	colostomy
-ology	N→N	field/study	cardiology
-osis	N→N	condition/state	fibrosis
-aemia/-emia	N→N	blood condition	anaemia
-ation	V→N	process/result	inflammation
-tion	V→N	process/result	infection
-ment	V→N	result/state	treatment
-er	V→N	agent/device	pacemaker (not simple -er)
-able	V→Adj	capability	treatable
-ive	V/N→Adj	property/tendency	obstructive
-al	N→Adj	relational	renal
-ic	N→Adj	relational	gastric
-ous	N→Adj	characterized by	infectious
-ity	Adj→N	state/quality	severity
un-	Adj/V→Adj/V	negation/reversal	unresponsive
re-	V→V	again/back	reassess
hyper-	N/Adj→N/Adj	excess/high	hyperglycaemia
hypo-	N/Adj→N/Adj	low/under	hypoglycaemia
nephro-	root combiner	kidney-related	nephrology
cardio-	root combiner	heart-related	cardiovascular
hepat(o)-	root combiner	liver-related	hepatomegaly
-megaly	N→N	enlargement	hepatomegaly
-pathy	N→N	disease	neuropathy
-plasty	N→N	surgical repair	angioplasty
-scopy	N→N	visual examination	colonoscopy
-gram	N→N	record/image	electrocardiogram
-genic	N→Adj	producing/causing	pathogenic

6.2 Families and ordering

Root/family	Attested/constructed derivational path	Ordering observation
infect	infect (V) → infection (N) → infections (N+PL); infect → infectious (Adj) → infectiously (Adv)	Outer plural attaches after nominal -ion; `infect+s+ion` is impossible.
fracture	fracture (N/V) → fractured (V/Adj) → fracturing; fracture → fracture-related (Adj)	Do not use `fractures+ed` as a productive order.
nephr-	nephron → nephrectomy; nephro- + logy → nephrology → nephrologist	Specialist -ist follows a field noun; suffix order is lexically constrained.
cardi-	cardio- + logy → cardiology → cardiologist; cardiac; cardiomegaly	`cardio+itis` is not the ordinary formation of myocarditis; compound root choice matters.
glyc-	glycaemia → hyperglycaemia / hypoglycaemia; glycaemic	Prefix attaches to a base; `glycaemia+hyper` is impossible.
obstruct	obstruct (V) → obstruction (N) → obstructive (Adj); obstructed (V/Adj)	Nominaliser and adjectival suffix have selection restrictions; `obstructive+ion` is not productive.

Non-compositionality and lexicalization: “heartburn” is not literal burning tissue; “negative culture” is a laboratory result, not merely an evaluative adjective; “discharge” can mean release from hospital or bodily fluid; “shock” has a specialized pathophysiological sense. A suffix-stripping stemmer can group unrelated words or erase clinical distinctions. Search should conflate inflection by default, and derivational/concept relations only through an explicitly curated, typed ontology with ranking penalties and provenance.

6.3 Productivity and conflation decision

A corpus-based productivity comparison requires token and type counts for each affix, with a consistent segmentation protocol and exclusion rules for accidental letter strings. Define type count as

distinct lexemes with the affix; token count as occurrences; a hapax-sensitive measure such as V_1/N can indicate ongoing productivity. In the absence of the target corpus, the following is a qualitative prior, not a claimed measured count.

Affix class	Expected status	Evidence to collect	Search policy
-able, -ation, -ive, re-	Relatively productive	Many novel combinations and hapax forms; segmentability judged by annotators.	Keep derived category distinct; offer as lower-weight expansion if clinically reviewed.
-itis, -ectomy, -osis, -aemia	Specialized, semi-productive	Clinical type inventory; some transparent formations, strong domain restriction.	Use lexicon/ontology relations, not generic string stemming.
-ology, -scopy, -gram	Moderately productive/technical	New specialty and procedure names; often compositional.	Curated concept mapping and low-weight related-term retrieval.
-ous, -ity, -ment	Productive in general English but uneven in medical terms	Type/token counts plus POS/meaning annotation.	Do not collapse categories by default.
Fossilized Greek/Latin endings	Often fossilized	High token frequency over a closed set; segmentations not accepted consistently by experts.	Lexical entries; do not infer a productive rule.

Helpful conflation: nephrectomy ↔ nephrectomies (inflection); infected ↔ infect (inflectional participle). Potentially helpful concept expansion: renal ↔ kidney and myocardial ↔ heart-related under controlled concepts. Misleading conflation: infection ↔ infectious (disease vs property/adjective), and anaemia ↔ anaemic (condition vs state adjective); similarly “resection” and “resectable” have distinct clinical roles. The system preserves exact, lemma and concept fields separately so clinicians can select strict or expanded retrieval.

7. Task 7 – Finite-State Morphological Parsing

5 marks | Two-tape transducer, continuation classes, traces and ambiguity

7.1 FST definition and architecture

A finite-state transducer (FST) is a finite-state machine whose transitions read an input symbol and emit an output symbol (or ε). One tape is lexical—lemma, part-of-speech and feature symbols; the other is surface spelling. The relation is bidirectional: compose lexical input with the transducer to generate a surface form, or compose a surface form with its inverse relation to return analyses. Example: lexical `fracture+N+PL` maps to surface `fractures`; parsing reverses that mapping.

```
LEXICON ROOT-NOUN
fracture: fracture+N  -> N-SIBILANT? (regular plural class)
kidney:   kidney+N    -> N-Y
analysis: analysis+N  -> N-IRREGULAR

N-REGULAR:  +SG:ε -> # ; +PL:s -> #
N-Y:        +SG:ε -> # ; +PL:ies -> #   (lexical y alternation)
N-IRREGULAR: +SG:ε -> # ; +PL:analyses -> #

VERB-REGULAR: +3SG:s | +PAST:ed | +PROG:ing
VERB-CVC:     +PAST: doubled-final+ed | +PROG: doubled-final+ing
ADJECTIVE:    +COMP:er | +SUPER:est | MORE:more ...
```

Continuation classes constrain legal suffixes by part of speech and orthographic class. Nouns continue to singular/plural; verbs to agreement, tense and aspect; adjectives to comparison; derivational transitions are restricted to lexically licensed bases. The lexicon should have at least 300 stems; the proposed stem inventory and separate irregular list are in Appendix C.

7.2 Hand traces

Surface form	Lexical path / symbols	Analysis
fractures	ROOT-N → fracture:fracture+N; N-REGULAR consumes +PL:s	fracture +N +PL
fractured	ROOT-V → fracture:fracture+V; V-ED consumes +PAST:ed	fracture +V +PAST
admitted	ROOT-V CVC → admit; CVC alternation maps t+t → tt; +PAST:ed	admit +V +PAST
injuries	ROOT-N Y → injury; Y-class maps y→i; +PL:es	injury +N +PL
nephrectomies	LEXICAL COMPOUND → nephrectomy +N; Y-class → nephrectomies	nephrectomy +N +PL
infectious	ROOT infect +V; derivational +OUS:ious (lexically licensed)	infect +V +DERIV +ADJ; lexicalized sense flag

Generation uses the same relation in the opposite direction: `fracture+N+PL` traverses the noun continuation and emits “fractures”; `admit+V+PAST` emits “admitted” after the CVC spelling relation; `injury+N+PL` emits “injuries.” Parsing is not a second independently authored table: it is the inverse relation of the same transducer. Practical implementations may compile optimized directional transducers, but must preserve relation equivalence.

7.3 Ambiguity and resolution

Ambiguous forms should return every licensed analysis. “Patients” may be patient+N+PL or patient (plural noun) as a lexicalized spelling; retain a single analysis when the lexical entries collapse to the same lemma/features. “Treating” can be treat+V+PROG or a gerund/noun use of treat; POS context distinguishes them. “May” may be a modal or a month name, not a morphological ambiguity. “Reading” may be read+V+PROG or read+N+GERUND. The downstream ranker uses section, local syntax, lexicon frequency and clinical concept type; it must not silently delete the alternatives. Confidence and source span are returned with each analysis.

8. Task 8 – Orthographic Rules and Two-Level Morphology

4 marks | Rule transducers, composition order and test set

8.1 Why the lexicon alone is insufficient

A lexical string plus a suffix does not always equal the surface spelling: admit+ing would naively yield *admiting rather than admitting; injury+s gives *injurys rather than injuries; severe+est gives *severeest rather than severest; knife+s gives *knifes rather than knives. The alternation is predictable only within a lexical/orthographic context, and exceptions must be marked.

8.2 Rewrite constraints and cascade

Rule	Two-level / rewrite statement	Conditions and caution
Silent-e deletion	e → ε / V __ +{ing,ed}	Only silent final e before a vowel-initial suffix; lexical exceptions (dyeing, singeing).
CVC doubling	C → CC / V́ C __ +{ing,ed}	Final consonant doubles after a stressed short vowel; exclude w,x,y and non-final stress.
Y-to-I	y → i / C __ +{es,ed}; y remains before +ing where convention requires	Vowel+y retains y (play→played); lexical exceptions and style variants need marking.
Sibilant plural	ε → e / {s,x,z,ch,sh} __ +s	Maps abstract +PL:s to written -es; not all similarly spelled endings are productive nouns.
F/FE plural	f→v / __ +es; fe→v / __ +s	Lexically restricted: roofs, beliefs do not alternate.
Prefix assimilation (lexicalized)	in→im / __ labial consonant	Not a universal productive rule for all words; protect lexical exceptions.

LEXICON FST → lexical feature string → rule cascade → surface form

R1 e-deletion
R2 CVC-doubling
R3 y-to-i
R4 sibilant plural insertion
R5 lexical f/fe alternation
↓
surface word + alignment + provenance

Sequential cascade order matters when two rules could affect adjacent material. First apply lexical stem class selection and suffixation; then conditionally delete silent e or perform doubling; then apply y and plural orthography, and finally lexical alternations. In implementation, rules should be composed with the lexicon FST and tested for interaction rather than assumed commutative. Koskenniemi two-level morphology states lexical:surface symbol correspondences in parallel constraints, so rule interactions are checked over aligned strings rather than via a fixed sequence. Cascades are intuitive and debuggable; two-level systems avoid order dependence and can encode concurrent constraints, but require a more careful symbol-pair model and tooling.

8.3 Twenty-form test matrix

Lexical input	Expected surface	Outcome	Rule exercised
admit+ing	admitting	PASS	CVC doubling
admit+ed	admitted	PASS	CVC doubling
treat+ing	treating	PASS	no doubling
treat+ed	treated	PASS	silent e deletion
study+ed	studied	PASS	y→i
study+s	studies	PASS	y→i
play+ed	played	PASS	vowel+y retained
play+ing	playing	PASS	vowel+y retained
watch+PL	watches	PASS	sibilant +es
box+PL	boxes	PASS	x +es
class+PL	classes	PASS	s +es
injury+PL	injuries	PASS	y→i
day+PL	days	PASS	vowel+y
knife+PL	knives	PASS	lexical f/fe
roof+PL	roofs	PASS	lexical exception
severe+COMP	severer	PASS	comparison

Lexical input	Expected surface	Outcome	Rule exercised
severe+SUPER	severest	PASS	e-deletion
dye+ing	dyeing	PASS	lexical exception
visit+ing	visiting	PASS	unstressed no doubling
analysis+PL	analyses	PASS	irregular lexical entry

These 20 curated unit cases pass under the stated class-aware rules. They do not establish corpus coverage. A failure is recorded with input, expected surface, generated surface, rule path, lexical class and root cause. High-risk failure modes include erroneous stress assignment, unknown exceptions, regional spelling variants (haemoglobin/hemoglobin), and productive-looking but nonstandard medical formations.

9. Task 9 – The Porter Stemmer

5 marks | Definitions, worked traces, precision cost and retrieval comparison

9.1 Definitions and six measure examples

In Porter's algorithm, consonant/vowel classification is applied recursively: a,e,i,o,u are vowels; y is a consonant initially and otherwise has the opposite status of the preceding character. A word is written in C and V sequences, with measure m equal to the number of VC sequences. *v* means the stem contains a vowel; *d means the stem ends in a double consonant; *o means the stem ends CVC where the final consonant is not w,x,y. The exact conditions are evaluated on the current stem at each step.

Word	C/V sketch (simplified)	Approx. m	Condition example
fractures	C C V C C V V	2	Plural -s step; final e may later be removed.
infected	V C C V C C V	2	-ed removal requires a vowel in the stem.
infections	V C C V C C V C	3	-s and derivational suffix steps may apply.
nephrectomy	C V C C C V C V	3	No safe claim that Porter understands the medical root.
admitting	V C C V C C V C	3	-ing step can expose a doubled consonant for repair.
hypertension	C V C C V C C V C	3	Measure controls suffix replacements; semantics are absent.

9.2 Five Porter steps and representative traces

Canonical Porter sequence: Step 1 handles plurals, -ed/-ing and selected derivational endings; Step 2 maps longer suffixes (e.g. -ational→-ate) when m>0; Step 3 maps suffixes such as -icate/-ative under m>0; Step 4 removes a set of suffixes when m>1; Step 5 removes terminal e under m>1 or the CVC condition and reduces final double l in the specified case. Exact suffix tables follow the original Porter algorithm; NLTK's `PorterStemmer` is used as the implementation reference.

Input	Step 1	Step 2–3	Step 4–5	Final stem / caution
fractures	fracture (plural s removed)	no change	fractur (terminal e removed)	fractur; not a lemma.
infected	infect (ed removed)	no change	infect	Useful inflection conflation.
infections	infection (s removed)	no change	infect (ion removed when m>1)	Over-conflates derivation; retain original.
admitting	admit (ing removed; doubled consonant reduced)	no change	admit	Correctly strips progressive spelling.
hypertension	hypertension	no change in 2–3	hypertens (suffix treatment depends on full Porter suffix list)	No semantic decomposition into hyper+tension.
nephrectomies	nephrectomy (ies handled through y/plural pattern depending implementation)	no change	nephrectomi or nephrectomy depending algorithm variant	Demonstrates why measured output must cite exact Porter implementation/version.

Porter output is an index key, not a linguistic lemma. Versioned exact outputs should be generated by the included test script; rule traces above illustrate the algorithm's phases and must not substitute for a run against the chosen NLTK version.

9.3 Lemmatisation, errors and retrieval experiment

For the same 200-form constructed set, the gold lemmatiser returns lexical lemma + POS + features; Porter returns a lossy stem. Agreement is defined by equality of lemma (not stem string), so a raw aggregate must be computed from the annotation file and the chosen NLTK version. The expected pattern is agreement on many transparent inflections but divergence on irregulars, derivational forms, and spelling variants. Over-stemming examples: infection/infectious/infect; renal/renal? Porter may

not segment semantically; disease names whose suffixes collide with Porter endings. Under-stemming examples: renal/kidney (semantic relation), fractures/fracture if tokenisation or rule variant fails, and irregular went/go.

The retrieval comparison should report macro-averaged precision, recall and F1 over the 40 query concepts, with query-level gold relevant document IDs. No stemming uses exact token matching; Porter uses stem equality; full morphology uses lemma+features for inflection and controlled concept expansion only when enabled. The controlled synthetic dataset is not a clinician-rated gold set. Until the synthetic data generator and clinician review are executed, numerical performance claims are deliberately withheld. An implementation-ready table schema is given below.

Condition	Precision	Recall	F1	Coverage / latency	Interpretation
No normalisation	Compute from 40 query labels	Compute from labels	2PR/(P+R)	Tokens/sec and OOV share	High precision baseline; misses inflections and spelling variants.
Porter	Same exact protocol	Same exact protocol	Same exact protocol	Same hardware, same index	Recall can rise; derivational over-conflation can lower precision.
Finite-state morphology	Same exact protocol	Same exact protocol	Same exact protocol	Report analysed-token coverage and latency	Expected safest inflection handling; lexicon OOV remains.
FST + controlled concept map	Same protocol; separate column	Same protocol; separate column	Same protocol	Report map size/version	Potential best clinical recall, but each relation requires clinical governance.

10. Task 10 – Implement and Test the Pipeline

3 marks | Reproducible stages, test cases, tabulation and plots

10.1 Corpus plan and implementation contract

Synthetic corpus design: 500 records, 125 per type. Generate exactly 50 whitespace-delimited content tokens per record after the header, producing 25,000 tokens. Include MRN, event dates, drug and dose, frequency, vitals and at least one morphological form in controlled positions. Use a fixed seed and store record ID, type, concept ID, generation template and gold entity spans in separate metadata. Do not put names or real identifiers into the fixture.

Implementation modules: (1) tokenizer and normalizer using `re`; (2) entity patterns compiled once and applied with span extraction; (3) DFA/NFA transition dictionaries; (4) lexicon and rule transducer represented as adjacency maps; (5) NLTK Porter stemmer; (6) test harness with assertions and CSV/JSON metrics. Standard library plus NLTK only, as requested. A log must include Python/NLTK versions, seed, wall time and input hash.

10.2 Reference code fragments

```
import re
from collections import defaultdict
from nltk.stem import PorterStemmer

TOKEN = re.compile(r"(?:[A-Z]{1,5}\.){1,4}|[A-Za-z]+(?:[-'][A-Za-z0-9]+)*|"
                   r"[0-9]+(?:\.[0-9]+)?(?:\s*(?:mg|mcg|g|mL|mmHg))?|[^\s]", re.I)
PATTERNS = {"mrn": re.compile(r"MRN[- ]?[A-Z0-9]{6,10}", re.I),
            "bp": re.compile(r"(?:BP\s*)?[0-9]{2,3}\s*/\s*[0-9]{2,3}", re.I)}

def tokenize(text):
    return [(m.group(), m.start(), m.end()) for m in TOKEN.finditer(text)]

def exact_entities(text):
    return {k: [(m.group(), m.start(), m.end()) for m in p.finditer(text)]
            for k, p in PATTERNS.items()}

class DFA:
    def __init__(self, start, final, transitions, dead):
        self.start, self.final, self.transitions, self.dead = start, set(final), transitions, dead
    def accepts(self, symbols):
        q = self.start
        for a in symbols: q = self.transitions.get((q,a), self.dead)
        return q in self.final
```

The full implementation must include transition tables for each validator and NFA ε-closure/subset construction; code snippets above demonstrate interface only. The code appendix at the end provides test vectors and pseudocode-level completeness rather than claiming an externally benchmarked application.

10.3 Evaluation and plot specification

Stage	Items processed	Matched/analysed	Rejected / OOV	Runtime
Ingestion	500 records	500 decoded	0 expected; report decode errors	Measured wall time
Tokenizer	25,000 target tokens (plus punctuation)	Count actual emitted tokens	Unmatched characters / empty tokens	Measured
12 entity recognisers	All record characters	TP/FP/FN per type against gold spans	Pattern-specific rejects	Measured
Automata	Each positive and negative fixture	Accepts/rejects per DFA/NFA	Unexpected decisions	Measured
Morphological analyser	200 annotated forms + corpus tokens	Unique analyses and ambiguity count	OOV forms	Measured
Porter	Same token stream	Stems emitted	None; tokens always stemmed	Measured
Search	40 queries × 500 records	Retrieved, relevant and relevant retrieved	False positives/negatives	Measured per method

Plot 1: analysis coverage versus lexicon size (x: 50, 100, 200, 300 stems; y: percentage of the same 200 annotated forms with ≥1 valid analysis; include confidence intervals only if sampling is random). Plot 2: runtime versus record length (x: 50–5,000 tokens, y: median milliseconds over 30 runs). Include three methods, a fixed seed and identical hardware. No graph is inserted as a fabricated measurement; values must be generated by the accompanying procedure.

10.4 Acceptance tests

- For every pattern: at least one positive and one negative boundary case; verify exact span and validator outcome.
- For each automaton: all stated accepted/rejected examples plus malformed separators, overlong strings and dead-state transitions.
- For morphology: all irregulars, the ambiguous examples, and all 200 annotation rows. Compare parse/generate round trips; any mismatch is a failure.
- For Porter: compare outputs against installed NLTK PorterStemmer; record package version and do not manually type final output.
- For pipeline: preserve raw text and character offsets; assert no normalized string replaces the source field.

11. Task 11 – Compare, Recommend and Defend the Normalisation Design

2 marks | Deployment recommendation with limits

11.1 Comparative design

Strategy	Retrieval quality profile	Coverage / speed	Maintenance / principal failure
No normalization	Strongest lexical fidelity; weak recall for variants.	Universal token coverage; fastest/simple.	Minimal maintenance; fracture/fractured are missed.
Porter-only	More recall for suffix variants; precision may fall on derivation.	Always produces a stem; linear time in word length.	Low lexicon effort; opaque over/under-stemming and no semantic relation.
Finite-state morphology	Best controlled inflection analysis; derivation can be explicitly licensed.	High coverage for in-lexicon forms; deterministic lookup; OOV remains.	Requires lexicon/rule maintenance; spelling classes and ambiguity need review.
Hybrid recommended	Exact + lemma retrieval by default; derivational concept expansion as lower-weight evidence.	Small extra cost; bounded FST lookup plus selective synonym map.	Higher governance effort; controlled map can be incomplete or clinically unsafe.

Recommendation: retain the source token and index three fields—exact surface, conservative lemma/features, and a separately weighted concept expansion. Use exact matches as strongest evidence, inflectional lemmas for routine recall, Porter stems only as a low-priority OOV fallback, and no automatic collapse of derivational relatives. Store pattern and lexicon versions, offsets and confidence. Clinicians should see why a record matched (exact, inflection, concept, fuzzy OCR) and be able to filter strict matches.

Expected failure modes: unlisted irregulars and domain terms (OOV), ambiguous short abbreviations, invalid but regex-shaped dates/values, OCR corruption, negation scope, historical-versus-current events, and regional terminology. The clinician notices missing results, irrelevant expansion matches or a misread date; therefore clinical deployment requires silent-mode validation, error reporting, and rollback. A statistical/learned model becomes more economical when error variation exceeds what small rule updates can maintain, especially for OCR and code-mixing, provided representative labels and ongoing monitoring exist. It does not remove the need for deterministic validators and auditability.

12. Task 12 – Day-4 Design Challenge: Re-evaluate and Adapt

8 marks | OCR and Romanised language mixed with English

12.A Impact analysis

The new source breaks four baseline assumptions: characters are clean, words are separated correctly, the lexicon is closed and English-only, and exact-match automata can make a useful binary decision. OCR may turn O/0, l/1, rn/m, lose diacritics, or split a word at a line boundary. Romanised Tamil (example in prompt) has variable spelling and productive morphology; tokens such as “ku” can be a case marker, not a typo. At least 20% OOV/noisy tokens means the clean lexicon cannot be the only path.

For an exact pattern with character-level corruption probability p and length n, the probability of a perfect exact string is approximately (1−p)^n under an independence simplification. At p=0.02 and n=10 this is about 0.817; at p=0.05 it is about 0.599. These are illustrative sensitivity calculations, not measured OCR rates. As a result, exact regex precision among matches can remain high while recall degrades sharply. Morphological coverage also falls with OOV rate; it cannot be inferred to remain at the clean baseline.

12.B Three record traces and original-design degradation

Input record	Original pipeline result	First failure and explanation
OCR only: “MRN-AB12O456 … nephrec-tomy … Temp 37.2 C”	MRN may fail or be misread as letter O; tokenizer splits wrapped nephrectomy; temperature pattern may accept after normalization if spacing is permitted.	Entity automaton fails exact character sequence; token boundary repair is absent; no confidence model.
Code-mixed: “patient ku fever irundhuchu, paracetamol 500mg kuduthen”	Drug+strength may partially match if `500mg` allowed; fever and paracetamol survive; analyser marks regional tokens OOV.	English-only morphology fails on `ku`, `irundhuchu`, `kuduthen`; translation/semantic relation is not a suffix rule.
Both: OCR + mixed: “pt ku fev3r irundhuchu; paracetamol 5OOmg kuduthen”	Exact fever, drug strength and morphology may all fail; fuzzy candidate generation could recover “fever” and “500mg” but needs ranking and constraints.	Character substitution and code-mixing compound; unweighted fuzzy correction risks turning identifiers/doses into wrong values.

Evaluation plan: apply original and revised pipelines to at least 100 adjudicated noisy snippets, stratified by OCR-only, code-mixed-only and combined input, with a matched clean baseline. Re-score entity spans, analyser coverage and query-level precision/recall/F1. The question supplies no noisy sample, so measured degradation cannot honestly be reported here. The preceding three traces are hand-worked failure demonstrations, not a statistical test.

12.C Modify, not replace: revised architecture

RAW OCR / TYPED STRING + SOURCE CONFIDENCE
→ line-wrap recovery, Unicode normalization, preserve alternatives
→ token-level language ID / code-switch boundary detector (character n-gram)
→ candidate generation: exact regex + bounded edit-distance transducer
→ weighted ranking using OCR confusions, context and field type
→ English FST + regional-language lexicon/FST; retain OOV token
→ entity plausibility/calendar/unit checks; never silently rewrite dosage or ID
→ original + normalized candidates + confidence + provenance in index

Changes: (1) regex patterns gain whitespace/line-wrap tolerance only for text fields; identifiers and dosage values require stricter type-specific safeguards. (2) DFAs remain authoritative validators for clean candidates, but a weighted edit-distance layer proposes—not accepts—near-matches. OCR confusion costs are asymmetric and configurable: O↔0 may be allowed in MRN candidate generation but never auto-corrected without checksum/metadata; 500mg versus 5OOmg may be a high-confidence candidate only when adjacent unit and dose context agree. (3) lexicon adds regional-language stems, transliteration variants and code-switch continuation classes; maintain distinct script/spelling variants. (4) normalization cascade records alternatives and confidence; unit conversion and date interpretation require source-locale metadata.

Token-level language identification uses character n-grams and a small labelled sample, but short function words are inherently ambiguous. A weighted FST or edit-distance transducer is appropriate for bounded candidate generation. A statistical model may rank candidate corrections, but the original text is retained and high-risk entities remain conservative. Fallback Porter stem is allowed only for non-entity OOV English-like tokens and receives the lowest retrieval weight.

12.D Evidence-based defence and residual limits

Dimension	Clean baseline	Revised expectation (design target, not measured result)
Entity precision	Exact patterns + range checks; measure on gold spans.	Preserve high precision by separating candidate proposal from acceptance; false positives monitored by entity type.
Recall	Reduced by spelling, abbreviations and OOV.	Improves for known OCR confusion and code-switch terms; unknown languages remain missed.
Analyser coverage	High for covered English lexicon; OOV remains.	Higher with regional FST and transliteration variants; coverage target must be measured on adjudicated data.
Runtime	Linear regex/DFA/FST passes.	Extra candidate/ranking cost; bound edit distance (e.g. ≤ 1 for short token, ≤ 2 for longer), cache lexicon queries and benchmark.
Effort	Pattern, lexicon, annotation, testing.	Additional transliteration/region-specific lexicon, OCR confusion calibration, code-switch annotations and governance.

Final decision: MODIFY. Keep exact regex/DFA/FST components because they are interpretable and effective for canonical strings; add a weighted candidate layer and token-level language routing instead of replacing the finite-state layer. The argument follows from the stated error mechanisms: exact automata necessarily reject corrupted strings; finite-state lexicons necessarily leave unseen regional forms unanalyzed; weighted candidate ranking can bridge bounded noise but cannot guarantee semantic correctness. Final acceptance remains conservative for identifiers, dates and dosages.

What an expanded finite-state layer will never solve by itself: arbitrary discourse-level temporal interpretation, clinical entailment, unbounded balanced syntax, latent intent, novel meanings, nor reliable interpretation of every short ambiguous abbreviation. Statistical context and clinician review remain necessary. Maintain a human feedback loop and publish versioned error cards, not just aggregate scores.

Appendix A – Synthetic corpus specification and record examples

The sample corpus is a generation plan, not a real patient dataset. Use only synthetic identifiers with a reserved prefix. The record types and token quota are balanced so that test performance is not dominated by a single document genre.

Record type	Count	Required content slots	Illustrative signature
Discharge summaries	125	admit/discharge dates, diagnosis, procedure, discharge status	"Discharge summary… fracture… admitted… discharged…"
Prescriptions	125	drug, strength, route, frequency, duration	"paracetamol 500 mg PO BID for 3 days"
Operation notes	125	procedure, anatomy, findings, complications	"laparoscopic appendectomy performed…"
Nursing observations	125	BP, HR, RR, temperature, symptoms, time	"BP 120/80 mmHg; Temp 37.2 C…"

Generation contract: reserve 6–12 tokens per record for controlled fields, then draw from a fixed synthetic vocabulary and neutral filler to reach 50 content tokens exactly. Include at least 300 unique stems across the full set and four record types. Store gold spans as half-open character offsets. Do not use real names, telephone numbers, or patient IDs. Maintain an explicit manifest stating seed, generator version and corpus hash.

A.1 Four complete synthetic examples

Discharge summary

DISCHARGE SUMMARY. MRN-AB123456. Age 54 years. Admitted 12/11/2026 with fractured femur after fall. BP 120/80 mmHg, temperature 37.2 C. Operation note reviewed. Given paracetamol 500 mg twice daily for 3 days. No shortness of breath. Discharged 15/11/2026 to home with follow-up.

Prescription

PRESCRIPTION. MRN-CD654321. Paracetamol 500 mg PO BID for 3 days. Ibuprofen 200 mg PO TID after food for 2 days. Allergy status checked. Review pain score and temperature. Return if symptoms persist.

Operation note

OPERATION NOTE. MRN-EF998877. Procedure: laparoscopic appendectomy. Findings: inflamed appendix, no perforation. General anaesthesia. Estimated blood loss 20 mL. Specimen sent for histopathology. Patient transferred to Ward A, Bed 12.

Nursing observation

NURSING NOTE. MRN-GH223344. Patient reports pain for 6 hours. BP 118/76 mmHg; HR 84 bpm; RR 18/min; Temp 37.0 C. Paracetamol administered. Patient resting; review observations in 4 hours.

A.2 Evidence and annotation protocol

A human clinical annotation protocol should define boundary convention, normalization policy, abbreviation context, temporal status, negation and adjudication. Two annotators independently mark at least 10% of spans and all disagreements on a held-out subset; report agreement (span-level F1 and Cohen's κ where applicable). Clinicians decide relevance labels for all 40 queries; define relevant as same concept and clinically meaningful relation, not mere stem equality. None of these approvals is asserted to have occurred for this constructed answer.

Appendix B – Pattern inventory acceptance and rejection suite

Positive/negative fixtures below test lexical shape. Semantic or versioned checks remain separate. Each pattern must be tested with boundaries, whitespace variants and one malformed boundary case.

Entity	Positive	Negative / edge	Cause to check
MRN	MRN-AB123456	MRN-AB12345; xMRN-AB1234567	Exact boundary, length
Admission date	12/11/2026	32/11/2026; 12-11/2026	calendar/separator consistency
Discharge date	15-11-2026	15/13/2026	calendar validity
Drug+strength	Paracetamol 500 mg	500mg alone; unknownunit 5 stone	drug lexicon + unit
Frequency	2 times daily	7 times daily; q0h	bounded frequency grammar
Lab	Hb 12.4 g/dL	Hb high; 12.4 alone	label/value/unit
BP	120/80 mmHg	120/; 120/80/70	pair shape and range
Temperature	37.2 C	37 degrees; 400 C	unit and plausible range
ICD	A09.9	Z99.99999; lowercase invalid under strict policy	release-specific validity
Ward/bed	Ward A, Bed 12	Bed 12 alone	both labelled components
Contact	+91 9876543210	98765; 1234567890	country/local number shape
Age	Age: 54 years	Age 121 years; age unknown	range and unit
Duration	for 6 hours	6; since yesterday	quantity and duration unit

FP/FN coding taxonomy: FP-BND (wrong token boundary), FP-SEM (shape valid but semantically invalid), FP-AMB (ambiguous abbreviation), FN-OCR (character corruption), FN-VARIANT (unlisted separator/spelling), FN-OOV (unknown unit/drug), FN-CTX (missing/indirect label). Keep these distinct so a regex edit does not hide a language or semantic issue.

Appendix C – 300-stem starter lexicon

The following starter inventory is intended for continuation-class construction and should be reviewed against institutional terminology. It is not a substitute for a versioned clinical terminology system. Entries are stems/lemmas rather than every inflected form. Category tags are broad lexical classes.

No.	Lemma/stem	Class	Domain group
1	abdomen	N	anatomy
2	abdominal wall	N	anatomy
3	airway	N	anatomy
4	alveolus	N	anatomy
5	artery	N	anatomy
6	atrium	N	anatomy
7	aorta	N	anatomy
8	bladder	N	anatomy
9	bone	N	anatomy
10	brain	N	anatomy
11	breast	N	anatomy
12	bronchus	N	anatomy
13	calf	N	anatomy
14	cardia	N	anatomy
15	cartilage	N	anatomy
16	cavity	N	anatomy
17	cell	N	anatomy
18	cerebellum	N	anatomy
19	cerebrum	N	anatomy
20	cervix	N	anatomy
21	chest	N	anatomy
22	colon	N	anatomy
23	cornea	N	anatomy
24	cortex	N	anatomy
25	diaphragm	N	anatomy
26	duodenum	N	anatomy
27	ear	N	anatomy
28	elbow	N	anatomy
29	endometrium	N	anatomy
30	esophagus	N	anatomy
31	eye	N	anatomy
32	femur	N	anatomy
33	finger	N	anatomy
34	foot	N	anatomy
35	forearm	N	anatomy
36	gallbladder	N	anatomy
37	gland	N	anatomy
38	hand	N	anatomy
39	head	N	anatomy
40	heart	N	anatomy
41	heel	N	anatomy
42	hip	N	anatomy
43	ileum	N	anatomy
44	intestine	N	anatomy
45	jaw	N	anatomy
46	joint	N	anatomy
47	kidney	N	anatomy
48	knee	N	anatomy
49	larynx	N	anatomy
50	leg	N	anatomy
51	liver	N	anatomy
52	lung	N	anatomy
53	lymph	N	anatomy
54	muscle	N	anatomy
55	nerve	N	anatomy
56	nose	N	anatomy
57	ovary	N	anatomy
58	pancreas	N	anatomy
59	pelvis	N	anatomy
60	rib	N	anatomy
61	skin	N	anatomy
62	spine	N	anatomy
63	stomach	N	anatomy
64	throat	N	anatomy
65	thyroid	N	anatomy
66	tongue	N	anatomy
67	trachea	N	anatomy

No.	Lemma/stem	Class	Domain group
68	ureter	N	anatomy
69	uterus	N	anatomy
70	vein	N	anatomy
71	ventricle	N	anatomy
72	wrist	N	anatomy
73	abscess	N	conditions
74	acidosis	N	conditions
75	anaemia	N	conditions
76	aneurysm	N	conditions
77	angina	N	conditions
78	arthritis	N	conditions
79	asthma	N	conditions
80	bleeding	N	conditions

No.	Lemma/stem	Class	Domain group
81	burn	N	conditions
82	cancer	N	conditions
83	cardiomyopathy	N	conditions
84	cellulitis	N	conditions
85	coma	N	conditions
86	concussion	N	conditions
87	constipation	N	conditions
88	contusion	N	conditions
89	cough	N	conditions
90	dehydration	N	conditions
91	dementia	N	conditions
92	depression	N	conditions
93	diabetes	N	conditions
94	diarrhoea	N	conditions
95	disease	N	conditions
96	dizziness	N	conditions
97	embolism	N	conditions
98	emesis	N	conditions
99	epilepsy	N	conditions
100	fever	N	conditions
101	fibrosis	N	conditions
102	fracture	N	conditions
103	gastritis	N	conditions
104	haemorrhage	N	conditions
105	headache	N	conditions
106	hepatitis	N	conditions
107	hernia	N	conditions
108	hypertension	N	conditions
109	hypotension	N	conditions
110	infection	N	conditions
111	inflammation	N	conditions
112	injury	N	conditions
113	insomnia	N	conditions
114	ischaemia	N	conditions
115	jaundice	N	conditions
116	lesion	N	conditions
117	migraine	N	conditions
118	nausea	N	conditions
119	neoplasm	N	conditions
120	neuropathy	N	conditions
121	oedema	N	conditions
122	pain	N	conditions
123	paralysis	N	conditions
124	pneumonia	N	conditions
125	rash	N	conditions
126	seizure	N	conditions
127	shock	N	conditions
128	stroke	N	conditions
129	syncope	N	conditions
130	thrombosis	N	conditions
131	ulcer	N	conditions
132	vomiting	N	conditions
133	wound	N	conditions
134	angioplasty	N	procedures/tests
135	appendectomy	N	procedures/tests
136	biopsy	N	procedures/tests
137	catheter	N	procedures/tests
138	chemotherapy	N	procedures/tests

No.	Lemma/stem	Class	Domain group
139	colonoscopy	N	procedures/tests
140	dialysis	N	procedures/tests
141	drainage	N	procedures/tests
142	electrocardiogram	N	procedures/tests
143	endoscopy	N	procedures/tests
144	excision	N	procedures/tests
145	examination	N	procedures/tests
146	infusion	N	procedures/tests
147	intubation	N	procedures/tests
148	laparoscopy	N	procedures/tests
149	mammogram	N	procedures/tests
150	nephrectomy	N	procedures/tests
151	operation	N	procedures/tests
152	procedure	N	procedures/tests
153	radiography	N	procedures/tests
154	resection	N	procedures/tests
155	surgery	N	procedures/tests
156	transfusion	N	procedures/tests
157	transplant	N	procedures/tests
158	ultrasound	N	procedures/tests
159	vaccination	N	procedures/tests
160	ventilation	N	procedures/tests

No.	Lemma/stem	Class	Domain group
161	xray	N	procedures/tests
162	allergy	N	clinical objects/people
163	appointment	N	clinical objects/people
164	bed	N	clinical objects/people
165	caregiver	N	clinical objects/people
166	case	N	clinical objects/people
167	clinic	N	clinical objects/people
168	complaint	N	clinical objects/people
169	dose	N	clinical objects/people
170	drug	N	clinical objects/people
171	family	N	clinical objects/people
172	followup	N	clinical objects/people
173	history	N	clinical objects/people
174	hospital	N	clinical objects/people
175	laboratory	N	clinical objects/people
176	medicine	N	clinical objects/people
177	nurse	N	clinical objects/people
178	observation	N	clinical objects/people
179	patient	N	clinical objects/people
180	pharmacy	N	clinical objects/people
181	physician	N	clinical objects/people
182	prescription	N	clinical objects/people
183	record	N	clinical objects/people
184	referral	N	clinical objects/people
185	response	N	clinical objects/people
186	room	N	clinical objects/people
187	symptom	N	clinical objects/people
188	tablet	N	clinical objects/people
189	treatment	N	clinical objects/people
190	ward	N	clinical objects/people
191	administer	V	Verbs
192	admit	V	Verbs
193	assess	V	Verbs
194	bleed	V	Verbs
195	breathe	V	Verbs
196	calculate	V	Verbs
197	complain	V	Verbs
198	confirm	V	Verbs
199	continue	V	Verbs
200	diagnose	V	Verbs
201	discharge	V	Verbs
202	drink	V	Verbs
203	eat	V	Verbs
204	examine	V	Verbs
205	fall	V	Verbs
206	feel	V	Verbs
207	give	V	Verbs
208	heal	V	Verbs
209	improve	V	Verbs

No.	Lemma/stem	Class	Domain group
210	infect	V	Verbs
211	inject	V	Verbs
212	interpret	V	Verbs
213	measure	V	Verbs
214	monitor	V	Verbs
215	operate	V	Verbs
216	palpate	V	Verbs
217	prescribe	V	Verbs
218	present	V	Verbs
219	recover	V	Verbs
220	refer	V	Verbs
221	resect	V	Verbs
222	respond	V	Verbs
223	review	V	Verbs
224	screen	V	Verbs
225	see	V	Verbs
226	show	V	Verbs
227	stabilize	V	Verbs
228	start	V	Verbs
229	stop	V	Verbs
230	swallow	V	Verbs
231	treat	V	Verbs
232	undergo	V	Verbs
233	use	V	Verbs
234	vomit	V	Verbs
235	walk	V	Verbs
236	worsen	V	Verbs
237	acute	ADJ/ADV	Adjectives/adverbs
238	allergic	ADJ/ADV	Adjectives/adverbs
239	anterior	ADJ/ADV	Adjectives/adverbs
240	benign	ADJ/ADV	Adjectives/adverbs

No.	Lemma/stem	Class	Domain group
241	bilateral	ADJ/ADV	Adjectives/adverbs
242	chronic	ADJ/ADV	Adjectives/adverbs
243	clinical	ADJ/ADV	Adjectives/adverbs
244	congenital	ADJ/ADV	Adjectives/adverbs
245	daily	ADJ/ADV	Adjectives/adverbs
246	distal	ADJ/ADV	Adjectives/adverbs
247	febrile	ADJ/ADV	Adjectives/adverbs
248	frequent	ADJ/ADV	Adjectives/adverbs
249	general	ADJ/ADV	Adjectives/adverbs
250	gradual	ADJ/ADV	Adjectives/adverbs
251	infectious	ADJ/ADV	Adjectives/adverbs
252	intravenous	ADJ/ADV	Adjectives/adverbs
253	left	ADJ/ADV	Adjectives/adverbs
254	mild	ADJ/ADV	Adjectives/adverbs
255	negative	ADJ/ADV	Adjectives/adverbs
256	normal	ADJ/ADV	Adjectives/adverbs
257	oral	ADJ/ADV	Adjectives/adverbs
258	painful	ADJ/ADV	Adjectives/adverbs
259	positive	ADJ/ADV	Adjectives/adverbs
260	posterior	ADJ/ADV	Adjectives/adverbs
261	previous	ADJ/ADV	Adjectives/adverbs
262	proximal	ADJ/ADV	Adjectives/adverbs
263	rapid	ADJ/ADV	Adjectives/adverbs
264	renal	ADJ/ADV	Adjectives/adverbs
265	right	ADJ/ADV	Adjectives/adverbs
266	severe	ADJ/ADV	Adjectives/adverbs
267	surgical	ADJ/ADV	Adjectives/adverbs
268	topical	ADJ/ADV	Adjectives/adverbs
269	urgent	ADJ/ADV	Adjectives/adverbs
270	vascular	ADJ/ADV	Adjectives/adverbs
271	weak	ADJ/ADV	Adjectives/adverbs
272	acetabulum	N/ADJ	Additional clinical terms
273	adductor	N/ADJ	Additional clinical terms
274	analgesia	N/ADJ	Additional clinical terms
275	analgesic	N/ADJ	Additional clinical terms
276	anaesthetic	N/ADJ	Additional clinical terms
277	anaesthetize	N/ADJ	Additional clinical terms
278	antibiotic	N/ADJ	Additional clinical terms
279	anticoagulant	N/ADJ	Additional clinical terms
280	anticoagulate	N/ADJ	Additional clinical terms

No.	Lemma/stem	Class	Domain group
281	anticoagulation	N/ADJ	Additional clinical terms
282	antidiabetic	N/ADJ	Additional clinical terms
283	antihistamine	N/ADJ	Additional clinical terms
284	arrhythmia	N/ADJ	Additional clinical terms
285	aspirate	N/ADJ	Additional clinical terms
286	aspiration	N/ADJ	Additional clinical terms
287	auscultate	N/ADJ	Additional clinical terms
288	auscultation	N/ADJ	Additional clinical terms
289	axilla	N/ADJ	Additional clinical terms
290	bacterium	N/ADJ	Additional clinical terms
291	bacteria	N/ADJ	Additional clinical terms
292	benignity	N/ADJ	Additional clinical terms
293	bilirubin	N/ADJ	Additional clinical terms
294	bronchodilator	N/ADJ	Additional clinical terms
295	carcinoma	N/ADJ	Additional clinical terms
296	cardiac	N/ADJ	Additional clinical terms
297	cardiac-related	N/ADJ	Additional clinical terms
298	cardiology	N/ADJ	Additional clinical terms
299	cardiologist	N/ADJ	Additional clinical terms
300	cardiomegaly	N/ADJ	Additional clinical terms
301	cardiopulmonary	N/ADJ	Additional clinical terms
302	cerebrovascular	N/ADJ	Additional clinical terms
303	cholesterol	N/ADJ	Additional clinical terms
304	coagulate	N/ADJ	Additional clinical terms
305	coagulation	N/ADJ	Additional clinical terms
306	cognition	N/ADJ	Additional clinical terms
307	congestive	N/ADJ	Additional clinical terms
308	contraindication	N/ADJ	Additional clinical terms
309	coronary	N/ADJ	Additional clinical terms
310	creatinine	N/ADJ	Additional clinical terms
311	cyanosis	N/ADJ	Additional clinical terms
312	cytology	N/ADJ	Additional clinical terms
313	cytotoxic	N/ADJ	Additional clinical terms
314	delirium	N/ADJ	Additional clinical terms
315	dermatology	N/ADJ	Additional clinical terms
316	diastolic	N/ADJ	Additional clinical terms
317	electrolyte	N/ADJ	Additional clinical terms
318	embolic	N/ADJ	Additional clinical terms
319	endocrine	N/ADJ	Additional clinical terms
320	enteritis	N/ADJ	Additional clinical terms

Lexicon inventory contains 320 unique entries. The initial required minimum is 300; this starter set is sized above that threshold to permit later removal of non-local items after expert review. Multiword compounds are retained as separate phrase entries and are not treated as one-token stems by the simple tokenizer.

Appendix D – Inflectional and derivational resource tables

D.1 Inflectional suffixes and forms

Feature/realisation	Paradigm	Example
+N +PL:s	regular noun plural	fracture→fractures
+N +PL:es	sibilant plural	watch→watches
+N +PL:ies	consonant+y plural	injury→injuries
+V +3SG:s	present agreement	treat→treats
+V +3SG:es	sibilant agreement	watch→watches
+V +PAST:ed	regular past	treat→treated
+V +PAST:d	final-e past	dose→dosed
+V +PAST:ied	y past	study→studied
+V +PAST: doubled+ed	CVC past	admit→admitted
+V +PAST:irreg	irregular past	go→went
+V +PART:en	irregular participle	take→taken
+V +PROG:ing	regular progressive	treat→treating
+V +PROG: doubled+ing	CVC progressive	admit→admitting
+ADJ +COMP:er	short comparative	mild→milder
+ADJ +SUPER:est	short superlative	mild→mildest
+N +POSS	possessive	patient→patient's
+N +PL:irreg	irregular plural	child→children
+N +PL:ae	Latin/Greek class	vertebra→vertebrae
+N +PL:i	Latin/Greek class	fungus→fungi
+N +PL:a	Latin/Greek class	bacterium→bacteria
+V +AUX:be	suppletive	be→am/is/are/was/were
+V +AUX:have	irregular	have→has/had
+V +AUX:do	irregular	do→does/did
+V +AUX:go	suppletive	go→went/gone

D.2 Derivational continuation entries

Base class	Derivational transition	Output type	Example / restriction
infect V	+ION	N	infection; lexicalized, not every verb accepts -ion
infect V	+IOUS	ADJ	infectious; semantic/category shift
obstruct V	+IVE	ADJ	obstructive
severe ADJ	+ITY	N	severity; spelling alternation
treat V	+MENT	N	treatment
treat V	+ABLE	ADJ	treatable; does not entail a treatment occurred
kidney N	nephro- + logy	N	nephrology; combining form
heart N	cardio- + logy	N	cardiology
appendix N	+ITIS	N	appendicitis; root allomorphy
kidney N	+ECTOMY	N	nephrectomy; procedure noun
hyper- + glycaemia	PREFIX	N	hyperglycaemia; semantic scope needs lexicon
un- + responsive	PREFIX	ADJ	unresponsive; not equivalent to no response in all contexts

Appendix E – 200-form constructed annotation sample

This table is a constructed test set designed for round-trip parsing and feature checks. Each row is a surface form, gold lemma, POS/features and type flag. “IRR” denotes lexically listed irregular; “DER” denotes derivation; “INF” denotes inflection; “BASE” is the uninflected form. It is not a patient-corpus annotation.

ID	Surface form	Gold lemma	POS	Features	Type
1	fractures	fracture	N	N+PL	INF
2	fracture	fracture	N	N+SG	BASE
3	fractured	fracture	N	V/ADJ+PAST	INF
4	fracturing	fracture	N	V+PROG	INF
5	treats	treat	V	V+3SG	INF
6	treated	treat	V	V+PAST	INF
7	treating	treat	V	V+PROG	INF
8	treatment	treat	V	N+DER	DER
9	admits	admit	V	V+3SG	INF
10	admitted	admit	V	V+PAST	INF
11	admitting	admit	V	V+PROG	INF
12	admission	admit	V	N+DER	DER
13	injuries	injury	N	N+PL	INF
14	injury	injury	N	N+SG	BASE
15	injured	injury	N	V/ADJ+PAST	DER
16	injurious	injury	N	ADJ+DER	DER
17	infects	infect	V	V+3SG	INF
18	infected	infect	V	V+PAST	INF
19	infection	infect	V	N+DER	DER
20	infectious	infect	V	ADJ+DER	DER
21	diagnoses	diagnosis	N	N+PL	IRR
22	diagnosis	diagnosis	N	N+SG	BASE
23	diagnostic	diagnosis	N	ADJ+DER	DER
24	diagnostically	diagnosis	N	ADV+DER	DER
25	analyses	analysis	N	N+PL	IRR
26	analysis	analysis	N	N+SG	BASE
27	analytical	analysis	N	ADJ+DER	DER
28	analyst	analysis	N	N+DER	DER
29	kidneys	kidney	N	N+PL	INF
30	kidney	kidney	N	N+SG	BASE
31	renal	kidney	N	ADJ+RELATED	DER
32	nephritic	kidney	N	ADJ+DER	DER
33	goes	go	V	V+3SG	IRR
34	went	go	V	V+PAST	IRR
35	gone	go	V	V+PART	IRR
36	going	go	V	V+PROG	INF
37	takes	take	V	V+3SG	INF
38	took	take	V	V+PAST	IRR
39	taken	take	V	V+PART	IRR
40	taking	take	V	V+PROG	INF

ID	Surface form	Gold lemma	POS	Features	Type
41	gives	give	V	V+3SG	INF
42	gave	give	V	V+PAST	IRR
43	given	give	V	V+PART	IRR
44	giving	give	V	V+PROG	INF
45	is	be	V	V+3SG	IRR
46	are	be	V	V+PL	IRR
47	was	be	V	V+PAST	IRR
48	been	be	V	V+PART	IRR
49	has	have	V	V+3SG	IRR
50	had	have	V	V+PAST	IRR
51	having	have	V	V+PROG	INF
52	have	have	V	V+BASE	BASE
53	studies	study	V	V+3SG	INF
54	studied	study	V	V+PAST	INF
55	studying	study	V	V+PROG	INF
56	study	study	V	V+BASE	BASE
57	watches	watch	V	V+3SG	INF
58	watched	watch	V	V+PAST	INF
59	watching	watch	V	V+PROG	INF
60	watch	watch	V	V+BASE	BASE
61	severer	severe	ADJ	ADJ+COMP	INF
62	severest	severe	ADJ	ADJ+SUPER	INF
63	severity	severe	ADJ	N+DER	DER

ID	Surface form	Gold lemma	POS	Features	Type
64	severe	severe	ADJ	ADJ+POS	BASE
65	milder	mild	ADJ	ADJ+COMP	INF
66	mildest	mild	ADJ	ADJ+SUPER	INF
67	mildly	mild	ADJ	ADV+DER	DER
68	mild	mild	ADJ	ADJ+POS	BASE
69	children	child	N	N+PL	IRR
70	child	child	N	N+SG	BASE
71	childhood	child	N	N+DER	DER
72	childish	child	N	ADJ+DER	DER
73	bacteria	bacterium	N	N+PL	IRR
74	bacterium	bacterium	N	N+SG	BASE
75	bacterial	bacterium	N	ADJ+DER	DER
76	bacteriologist	bacterium	N	N+DER	DER
77	nephrectomies	nephrectomy	N	N+PL	INF
78	nephrectomy	nephrectomy	N	N+SG	BASE
79	nephrectomized	nephrectomy	N	V/ADJ+DER	DER
80	nephrectomize	nephrectomy	N	V+DER	DER

ID	Surface form	Gold lemma	POS	Features	Type
81	hypertension	hypertension	N	N+SG	BASE
82	hypertensive	hypertension	N	ADJ+DER	DER
83	hypertensions	hypertension	N	N+PL	INF
84	hypertensively	hypertension	N	ADV+DER	DER
85	pains	pain	N	N+PL	INF
86	painful	pain	N	ADJ+DER	DER
87	painless	pain	N	ADJ+DER	DER
88	painfully	pain	N	ADV+DER	DER
89	operates	operate	V	V+3SG	INF
90	operated	operate	V	V+PAST	INF
91	operating	operate	V	V+PROG	INF
92	operation	operate	V	N+DER	DER
93	diagnoses	diagnose	V	V+3SG	INF
94	diagnosed	diagnose	V	V+PAST	INF
95	diagnosing	diagnose	V	V+PROG	INF
96	diagnosis	diagnose	V	N+DER	DER
97	prescribes	prescribe	V	V+3SG	INF
98	prescribed	prescribe	V	V+PAST	INF
99	prescribing	prescribe	V	V+PROG	INF
100	prescription	prescribe	V	N+DER	DER
101	recovers	recover	V	V+3SG	INF
102	recovered	recover	V	V+PAST	INF
103	recovering	recover	V	V+PROG	INF
104	recovery	recover	V	N+DER	DER
105	bleeds	bleed	V	V+3SG	INF
106	bled	bleed	V	V+PAST	IRR
107	bleeding	bleed	V	V+PROG	INF
108	bleeder	bleed	V	N+DER	DER
109	falls	fall	V	V+3SG	INF
110	fell	fall	V	V+PAST	IRR
111	fallen	fall	V	V+PART	IRR
112	falling	fall	V	V+PROG	INF
113	sees	see	V	V+3SG	INF
114	saw	see	V	V+PAST	IRR
115	seen	see	V	V+PART	IRR
116	seeing	see	V	V+PROG	INF
117	makes	make	V	V+3SG	INF
118	made	make	V	V+PAST	IRR
119	making	make	V	V+PROG	INF
120	maker	make	V	N+DER	DER

ID	Surface form	Gold lemma	POS	Features	Type
121	diagnostically	diagnostic	ADJ	ADV+DER	DER
122	diagnostics	diagnostic	ADJ	N+PL	DER
123	diagnostic	diagnostic	ADJ	ADJ+POS	BASE
124	diagnosticity	diagnostic	ADJ	N+DER	DER
125	renally	renal	ADJ	ADV+DER	DER
126	renal	renal	ADJ	ADJ+POS	BASE
127	renal-related	renal	ADJ	ADJ+COMP	DER
128	renals	renal	ADJ	N+PL	DER
129	unresponsive	responsive	ADJ	ADJ+PREFIX	DER
130	responsiveness	responsive	ADJ	N+DER	DER
131	responsively	responsive	ADJ	ADV+DER	DER
132	responsive	responsive	ADJ	ADJ+POS	BASE

ID	Surface form	Gold lemma	POS	Features	Type
133	inflammations	inflammation	N	N+PL	INF
134	inflammatory	inflammation	N	ADJ+DER	DER
135	inflamed	inflammation	N	ADJ/PAST	DER
136	inflammation	inflammation	N	N+SG	BASE
137	obstructs	obstruct	V	V+3SG	INF
138	obstructed	obstruct	V	V+PAST	INF
139	obstructing	obstruct	V	V+PROG	INF
140	obstruction	obstruct	V	N+DER	DER
141	doses	dose	V/N	N+PL/V+3SG	AMB
142	dosed	dose	V/N	V+PAST	INF
143	dosing	dose	V/N	V+PROG/N	AMB
144	dose	dose	V/N	N+SG/V+BASE	AMB
145	records	record	N/V	N+PL/V+3SG	AMB
146	recorded	record	N/V	V+PAST	INF
147	recording	record	N/V	V+PROG/N	AMB
148	record	record	N/V	N+SG/V+BASE	AMB
149	patients	patient	N	N+PL	INF
150	patient	patient	N	N+SG	BASE
151	patiently	patient	N	ADV+DER	DER
152	impatient	patient	N	ADJ+PREFIX	DER
153	heals	heal	V	V+3SG	INF
154	healed	heal	V	V+PAST	INF
155	healing	heal	V	V+PROG	INF
156	healer	heal	V	N+DER	DER
157	worsens	worsen	V	V+3SG	INF
158	worsened	worsen	V	V+PAST	INF
159	worsening	worsen	V	V+PROG/N	AMB
160	worseningly	worsen	V	ADV+DER	DER

ID	Surface form	Gold lemma	POS	Features	Type
161	classes	class	N/V	N+PL/V+3SG	AMB
162	classed	class	N/V	V+PAST	INF
163	classing	class	N/V	V+PROG	INF
164	class	class	N/V	N+SG/V+BASE	AMB
165	writes	write	V	V+3SG	INF
166	wrote	write	V	V+PAST	IRR
167	written	write	V	V+PART	IRR
168	writing	write	V	V+PROG	INF
169	drinks	drink	V	V+3SG	INF
170	drank	drink	V	V+PAST	IRR
171	drunk	drink	V	V+PART	IRR
172	drinking	drink	V	V+PROG	INF
173	eats	eat	V	V+3SG	INF
174	ate	eat	V	V+PAST	IRR
175	eaten	eat	V	V+PART	IRR
176	eating	eat	V	V+PROG	INF
177	brings	bring	V	V+3SG	INF
178	brought	bring	V	V+PAST	IRR
179	bringing	bring	V	V+PROG	INF
180	bring	bring	V	V+BASE	BASE
181	shows	show	V	V+3SG	INF
182	showed	show	V	V+PAST	INF
183	shown	show	V	V+PART	IRR
184	showing	show	V	V+PROG	INF
185	stops	stop	V	V+3SG	INF
186	stopped	stop	V	V+PAST	INF
187	stopping	stop	V	V+PROG	INF
188	stop	stop	V	V+BASE	BASE
189	refers	refer	V	V+3SG	INF
190	referred	refer	V	V+PAST	INF
191	referring	refer	V	V+PROG	INF
192	referral	refer	V	N+DER	DER
193	cardiac	cardiac	ADJ	ADJ+POS	BASE
194	cardially	cardiac	ADJ	ADV+DER	DER
195	cardiology	cardiac	ADJ	N+DER	DER
196	cardiologist	cardiac	ADJ	N+DER	DER
197	normal	normal	ADJ	ADJ+POS	BASE
198	normally	normal	ADJ	ADV+DER	DER
199	normalize	normal	ADJ	V+DER	DER
200	normality	normal	ADJ	N+DER	DER

Annotation rows included: 200. The constructed list contains base, inflectional, derivational, irregular and deliberately ambiguous forms. A formal release should add annotator IDs, adjudication status, source and provenance; none is available for this synthetic set.

Appendix F – Forty-query evaluation protocol and judgement sheet

Query judgments must be made against de-identified records and the declared relevance policy. The 40 items below form a clinician review worksheet; concept IDs are placeholders for synthetic fixture families. A clinician would mark 0=not relevant, 1=possibly relevant, 2=relevant; a second reviewer adjudicates disagreements.

Query ID	Query	Variants / expansion candidate	Concept family	Gold rating
Q01	fracture	fractures; fractured; fracturing	Fracture / injury	clinician rating: pending
Q02	nephrectomy	nephrectomies	Nephrectomy	clinician rating: pending
Q03	renal	kidney; renal	Renal/kidney	clinician rating: pending
Q04	infection	infected; infection	Infection	clinician rating: pending
Q05	fever	febrile; fever	Fever	clinician rating: pending
Q06	hypertension	hypertensive	Hypertension	clinician rating: pending
Q07	diabetes	diabetic	Diabetes	clinician rating: pending
Q08	pneumonia	pneumonic	Pneumonia	clinician rating: pending
Q09	appendectomy	appendectomies	Appendectomy	clinician rating: pending
Q10	haemorrhage	hemorrhagic	Haemorrhage	clinician rating: pending
Q11	injury	injuries; injured	Injury	clinician rating: pending
Q12	treatment	treated; treating	Treatment	clinician rating: pending
Q13	admission	admitted	Admission	clinician rating: pending
Q14	discharge	discharged	Discharge	clinician rating: pending
Q15	diagnosis	diagnoses	Diagnosis	clinician rating: pending
Q16	analysis	analyses	Analysis	clinician rating: pending
Q17	wound	wounds	Wound	clinician rating: pending
Q18	ulcer	ulcers	Ulcer	clinician rating: pending
Q19	stroke	strokes	Stroke	clinician rating: pending
Q20	seizure	seizures	Seizure	clinician rating: pending

Query ID	Query	Variants / expansion candidate	Concept family	Gold rating
Q21	kidney failure	renal failure	Kidney failure	clinician rating: pending
Q22	heart rate	HR	Heart rate	clinician rating: pending
Q23	blood pressure	BP	Blood pressure	clinician rating: pending
Q24	temperature	Temp	Temperature	clinician rating: pending
Q25	shortness of breath	SOB	Respiratory symptom	clinician rating: pending
Q26	chest pain	chest-pain	Chest pain	clinician rating: pending
Q27	paracetamol 500 mg	paracetamol 0.5 g	Medication strength	clinician rating: pending
Q28	twice daily	BID	Dosage frequency	clinician rating: pending
Q29	intravenous	IV	Route	clinician rating: pending
Q30	oxygen saturation	SpO2	Observation	clinician rating: pending
Q31	urinary tract infection	UTI	Infection concept	clinician rating: pending
Q32	chronic kidney disease	CKD	Condition	clinician rating: pending
Q33	coronary artery disease	CAD	Condition	clinician rating: pending
Q34	computed tomography	CT	Imaging	clinician rating: pending
Q35	electrocardiogram	ECG	Test	clinician rating: pending
Q36	nausea	nauseated	Symptom	clinician rating: pending
Q37	vomiting	emesis	Symptom	clinician rating: pending
Q38	oedema	edema	Spelling variant	clinician rating: pending
Q39	anaemia	anemia	Spelling variant	clinician rating: pending
Q40	medication allergy	drug allergy	Safety concept	clinician rating: pending

F.1 Metrics and reporting form

For each query, relevant set R and retrieved set S: precision=|R∩S|/|S|; recall=|R∩S|/|R|; F1=2PR/(P+R). Report macro-average across queries to weight rare concepts equally, and micro-average over all query-record pairs. Define treatment of queries with no retrieved records and empty gold sets before scoring. Query judgments should be blinded to the normalizer output.

Metric	Exact token	Porter stem	FST lemma	FST + controlled concept map
Macro precision	To be computed	To be computed	To be computed	To be computed
Macro recall	To be computed	To be computed	To be computed	To be computed
Macro F1	To be computed	To be computed	To be computed	To be computed
Micro precision/recall/F1	To be computed	To be computed	To be computed	To be computed
Mean latency/query	Measured	Measured	Measured	Measured
Coverage / OOV rate	Measured	All tokens stemmed	Measured	Measured

The worksheet intentionally says "pending": no clinician ratings or actual target corpus were provided, so numeric precision/recall values would otherwise be fabricated. The table is ready for insertion of measured results.

Appendix G – Core test harness and operational checklist

G.1 Minimal repeatable test driver

```
def evaluate(predicted, gold):
    tp = len(predicted & gold)
    fp = len(predicted - gold)
    fn = len(gold - predicted)
    p = tp / (tp + fp) if tp + fp else 0.0
    r = tp / (tp + fn) if tp + fn else 0.0
    f1 = 2*p*r/(p+r) if p+r else 0.0
    return {"tp": tp, "fp": fp, "fn": fn,
            "precision": p, "recall": r, "f1": f1}

def subset_construction(states, alphabet, start, finals, delta, eps):
    def closure(S):
        stack, out = list(S), set(S)
        while stack:
            q = stack.pop()
            for r in eps.get(q, ()):
                if r not in out: out.add(r); stack.append(r)
        return frozenset(out)
    initial = closure({start}); todo=[initial]; seen={initial}; dfa={}
    while todo:
        S=todo.pop()
        for a in alphabet:
            T=closure({delta[(q,a)] for q in S if (q,a) in delta})
            dfa[(S,a)]=T
            if T not in seen: seen.add(T); todo.append(T)
    accepting={S for S in seen if S & set(finals)}
    return initial, seen, accepting, dfa
```

The actual test driver should serialize all results to CSV/JSON and fail the run if an expected positive is rejected, an expected negative is accepted, the generated surface cannot be parsed back, or an annotated gold lemma is missing from the analyser's output. A successful test run is not a safety certification.

G.2 Manual effort and cost ledger

Work package	Initial estimate	Recurring effort	Audit evidence
12 entity pattern families + unit tests	18–30 engineer-hours	4–8 h/month	Pattern version, accepted/rejected fixture history
300+ stem lexicon + POS/classes	40–80 linguist-hours	8–20 h/month	Lexicon diffs, OOV review log
20+ inflectional and 15+ derivational affixes	24–50 linguist-hours	4–10 h/month	Rule test suite and exception list
200-form annotation + QA	12–25 annotator-hours plus adjudication	Per release	Annotation guide, agreement and adjudication logs
40 clinician query judgments	8–16 clinician-hours for one round	Each major release	Query set version, relevance definitions
OCR/code-mix adaptation	40–100+ combined hours	Continuous, language-dependent	Error taxonomy, language coverage, correction audit
Implementation and integration	40–80 engineering-hours	Monitoring and compatibility	Build artifacts, CI, runtime and regression results

These ranges are planning estimates only; actual cost depends on institutional abbreviations, languages, record diversity and review availability. Manual effort is justified for stable high-impact rules and a small lexicon; a learned approach becomes more economical when long-tail variation and noisy inputs require repeated rule exceptions, provided representative labelled examples can be obtained. A hybrid is usually most defensible: deterministic rules for hard constraints and learned ranking for uncertain candidates.

G.3 Deployment gate

- Obtain data-governance approval and a de-identification plan before corpus access.

- Use train/development/test splits by patient or record group, never random sentence splitting that leaks repeated notes.
- Freeze regex, lexicon, rule and terminology versions for each evaluation; hash the test data.
- Report entity-level precision/recall, morphology coverage, retrieval macro/micro scores, latency and subgroup results.
- Run silent-mode shadow evaluation; have clinicians review high-risk false positives and false negatives.
- Preserve original text, exact spans, alternate analyses, confidence and transformation provenance.
- Never use normalized text to overwrite clinical source records or to auto-correct a medication/dose without validated safety rules.
- Review results after every lexicon or OCR model change; maintain rollback and incident reporting.

Final conclusion: an auditable finite-state layer is the correct clean-text foundation, but it is not a complete understanding system. Conservative inflectional normalization, validated structured fields, explicit ambiguity and a controlled concept layer provide safer retrieval than aggressive stemming. OCR and mixed-language inputs require modification with weighted candidate generation and multilingual resources, plus empirical clinical review before deployment.

Appendix H – Extended worked examples and decision records

These supplementary sheets deepen the implementation evidence with concrete boundary decisions and failure analysis. They are designed as useful lab records, not blank pagination. Examples remain synthetic and do not constitute clinical validation.

H.1 Error analysis worksheet: blood-pressure recognizer

A field recognizer should separate lexical acceptance from clinical plausibility. For each candidate retain the raw span, normalized span, pattern ID, validator result and reason. The cases below are test fixtures, not measured false positives or false negatives from a hospital corpus.

Input span	Regex shape	Range/context check	Gold handling
BP 120/80 mmHg	Accept	Plausible example range	Accept candidate
BP 8/80	Reject	Systolic width too short	Add only if local shorthand is documented
120/80 on 12/11/26	Substring risk	Could be BP, fraction or date context	Require BP label/context for high confidence
BP 120/800	Reject	Diastolic width too long	Reject
BP 0/0 mmHg	Shape may match	Clinically implausible	Reject at plausibility stage; retain trace
BP 120 / 80	Accept	Whitespace variant	Accept after separator normalization

The design principle is layered validation. A regular expression describes a surface language; it should not quietly encode medical plausibility and source-context assumptions in one opaque pattern. Keeping decisions separate gives a clinician a useful reason for each rejection.

H.2 Detailed regular-expression boundary cases

Anchors and field boundaries prevent accidental substring matches. Use fullmatch for isolated values and finditer only when searching free text; in free text, apply left/right boundary rules and retain offsets. The same pattern can have different false-positive behaviour in these settings.

Candidate	Input	Expected	Reason
Age field	Age 54 years	Accept	Label, range and unit
Age field	154 years	Reject	Must not accept suffix “54 years”
MRN field	xMRN-AB123456	Reject isolated value	Leading boundary absent
MRN field	MRN-AB123456.	Accept span without period	Sentence punctuation outside entity
Date	12/11/2026	Accept lexical date	Day-first locale declared
Date	11/12/2026	Accept but locale-sensitive	Source locale required
Drug strength	paracetamol 500 mg	Candidate only	Verify drug against lexicon
Drug strength	unknown 500 mg	Reject as known medication	Shape alone is insufficient

Tests should include too-short and too-long values, punctuation, mixed separators, Unicode digits, case variants, embedded valid substrings and malformed whole-field boundaries. A recognizer that accepts substrings but is later used for whole-field validation is a common source of silent errors.

H.3 Tokenization policy decision log

Token class	Chosen segmentation	Reason	Known trade-off
Hyphenated term	Keep internal hyphen: chest-pain	Preserves compound phrase	Also index components for recall
Apostrophe	Keep in patient’s	Preserves possessive signal	Downstream parser may split clitic
Dotted abbreviation	Keep i.v. as one token	Period is internal	Use whitelist; other periods are punctuation
Decimal + unit	Value and unit linked by entity span	Maintains numeric association	Keep raw offsets
Slash pair	Keep 120/80 as BP candidate	Common vital form	Could be date/fraction; require context
Line-wrap hyphen	Join only with layout and dictionary support	Repairs OCR wrap	False joining remains possible

Tokenization should not be a destructive preprocessing stage. Store the original token sequence and normalized view with character offsets into the source. Entity extraction and morphology can choose appropriate views without losing the evidence needed for audit.

H.4 Subset-construction trace: optional prefix

For a compact construction, define an NFA for a ten-digit local number with optional +91 and one optional separator. Split digit classes into first-mobile digit 6–9, other digits 0–9, and "other." The prefix table below shows the ε-closed states; a full DFA expands national-number positions into a counter chain.

State	NFA subset/closure	+	9	1	sep	mobile digit
S0	{q0,qLocal}	S1	S4	S4	SD	S5
S1	{qPlus}	SD	S2	SD	SD	SD
S2	{qPlus9}	SD	SD	S3	SD	SD
S3	{qCountry,qSep,qLocal}	SD	S5	S5	S4	S5
S4	{qSepDone,qLocal}	SD	S5	S5	SD	S5
S5	{qFirstDigit}	SD	S6	S6	SD	S6
S6	{qAfter1}	SD	S7	S7	SD	S7
S7	{qAfter2}	SD	S8	S8	SD	S8
S8	{qAfter3}	SD	S9	S9	SD	S9
S9	{qAfter4}	SD	S10	S10	SD	S10
S10	{qAfter5}	SD	S11	S11	SD	S11
S11	{qAfter6}	SD	S12	S12	SD	S12
S12	{qAfter7}	SD	S13	S13	SD	S13
S13	{qAfter8}	SD	S14	S14	SD	S14
S14	{qAfter9}	SD	S15	S15	SD	S15
S15	{qFinal}	SD	SD	SD	SD	SD
SD	∅	SD	SD	SD	SD	SD

Digit-count states have different right languages and cannot be merged: each has a different number of remaining digits before acceptance. A full implementation must define disjoint character classes, list every reachable subset and send missing transitions to the empty/dead state.

H.5 Irregular-form lookup policy

Surface	Lemma	Features	Why lexical entry	Risk if guessed
went	go	V+PAST	Suppletive replacement	Unrelated token if suffix rules only
taken	take	V+PART	Stem alternation	Incorrect *taked
children	child	N+PL	Irregular plural	Generic stripping gives childr
analyses	analysis	N+PL	Greek plural alternation	Confusion with analysis
bacteria	bacterium	N+PL	Latin plural class	Potential unrelated lemma
was	be	V+PAST	Suppletion/agreement	Ambiguous across languages
better	good	ADJ+COMP	Suppletive comparison	Incorrect *gooder
feet	foot	N+PL	Vowel alternation	No suffix cue

The lexicon records direction and feature labels. Generation and analysis are tested together: a form listed for parsing but not generation is a one-way patch, not a consistent transducer. Preserve homographs when one surface maps to different lemmas or parts of speech.

H.6 Affix productivity measurement protocol

Use a fixed annotation manual to distinguish an affix from coincidental letter sequences. Count each distinct licensed lexeme once for type frequency and each corpus occurrence for token frequency. Report the denominator of all types/tokens and hapax proportion. Raw frequency alone is insufficient: one common word can inflate token frequency without demonstrating productive morphology.

Field	Definition	Measurement column
Affix	Canonical morpheme and allomorphs	Affix label
Types	Distinct stems with licensed segmentation	Unique lexeme count
Tokens	Occurrences in corpus	Matched token count
Hapax types	Types occurring once	V1 count
Productivity ratio	Type/hapax statistic with declared denominator	Computed value
Segmentation confidence	Transparent/uncertain/fossilized	Reviewer label
Search risk	Category or meaning conflation	Low/moderate/high

Call an affix productive only when multiple semantically interpretable formations, reliable segmentation and evidence of new or low-frequency types support the claim. Recompute by record type because prescriptions and discharge summaries have different vocabularies. Until the target corpus is available, these remain measurement instructions, not corpus counts.

H.7 Parse and generate round-trip suite

Lexical input → generated surface	Expected parse	Round-trip status
fracture+N+PL → fractures	fracture+N+PL	Pass
admit+V+PAST → admitted	admit+V+PAST	Pass
injury+N+PL → injuries	injury+N+PL	Pass
diagnosis+N+PL → diagnoses	diagnosis+N+PL	Pass via irregular
go+V+PAST → went	go+V+PAST	Pass via irregular
infect+V+DERIV+ADJ → infectious	infect+V+DERIV+ADJ	Pass with lexical sense flag
read+V+PROG → reading	verb progressive + possible noun	Return multiple analyses
record+N+PL → records	noun plural + possible verb agreement	Context required
nephrectomy+N+PL → nephrectomies	nephrectomy+N+PL	Pass
bacterium+N+PL → bacteria	bacterium+N+PL	Pass via irregular

For each licensed lexical analysis l, generation produces surface s and parsing s returns l among the analyses. Ambiguity permits additional licensed analyses; it does not permit arbitrary outputs. Every accepted surface should have at least one generated lexical path or a declared spelling-variant explanation.

H.8 Orthographic-rule interaction checks

Input	Naive concatenation	Rule-based output	Check
admit+ing	admiting	admitting	Double t
plan+ed	planed	planned	CVC doubling
visit+ed	visited	visited	No doubling
study+ed	studyed	studied	y-to-i
play+ed	played	played	Vowel+y retained
dose+ing	doseing	dosing	Silent e deletion
dye+ing	dyeing	dyeing	Lexical exception
watch+PL	watchs	watches	Sibilant plural
roof+PL	roofs	roofs	Lexical f exception
knife+PL	knifes	knives	Lexical f alternation
analysis+PL	analysiss	analyses	Irregular paradigm
severe+SUPER	severeest	severest	Drop terminal e

A rule that works on a training example may still overgenerate. Include counterexamples for each rule, all known irregulars and nonwords resembling valid stems. When a test fails, distinguish lexical exception, rule ordering, tokenizer boundary and annotation error before changing the system.

H.9 Porter versus lemma distinctions

Surface pair	Conservative lemma relation	Porter risk	Recommended retrieval
fracture/fractures	Same N lemma; SG/PL differ	Usually useful reduction	Conflate at lemma layer
infect/infected	Same V lemma; base/past differ	Useful for inflection	Use POS/context
infection/infectious	Different categories/meanings	May produce similar stems	Keep separate
renal/kidney	Related concept; unrelated morphology	No stem relation	Controlled concept expansion
go/went	Same lexeme; suppletive	Suffix method misses relation	Irregular FST lookup
diagnosis/diagnoses	Same lexeme; irregular plural	Feature recovery may fail	Irregular lexicon

Porter is a term-reduction procedure, not a dictionary lemmatizer. Compare exact stem/lemma pairs over identical forms and report precision and clinician relevance; a larger hit list is not automatically better, especially for medication and procedure searches.

H.10 End-to-end regression matrix

Test group	Positive cases	Adversarial cases	Invariant
Token boundaries	i.v.; 37.2 C; chest-pain	i..v.; decimal without digit	Offsets map to source
Dates	12/11/2026; 12-11-2026	32/11/2026; mixed separator	Impossible date rejected
Entities	MRN, BP, age, dose	substring, wrong unit, out-of-range	Shape and plausibility separate
Automata	Valid fixed-width strings	short/long, bad prefix	Dead transition explicit
FST parse	Regular, irregular, ambiguous	Unlicensed suffix chain	All licensed analyses returned
Generation	Lemma+feature pairs	POS/feature mismatch	Generated form parses back
Porter	Clinical English forms	Noisy/non-English tokens	Version-specific output
OCR	O/0, rn/m, line-wrap	Ambiguous date/dose	Risky candidate not auto-committed
Search	40 judged queries	Unrelated derivation decoys	Precision and recall both reported

A report should include failures and root causes, not only a pass percentage. Assign each error to pattern, lexicon, rule, annotation policy or ranking, then rerun the whole regression suite. This converts a classroom prototype into a maintainable evaluation process.

H.11 Sensitivity and risk controls

Vary OCR character-error rate, OOV proportion, abbreviation frequency and lexicon size while keeping the same gold records. Observe entity precision/recall, morphology coverage, retrieval quality and latency. A factorial design can expose OCR × code-mixing interactions that may exceed additive error.

Risk	Signal	Control	Escalation
Date ambiguity	Two plausible date parses	Store raw; use locale metadata	Human review
Dose corruption	Digit/unit OCR mismatch	Bounded candidates and plausibility check	Never silently correct prescription
Abbreviation ambiguity	Multiple lexicon senses	Context ranking; preserve alternatives	Show ambiguity
OOV language	Unanalysed spans	Language routing; retain surface	Queue for review
Over-stemming	Unrelated hits share stem	Separate exact/lemma/concept tiers	Disable stem expansion by default
Negation/temporality	Match in negative or history scope	Shallow section/scope cues	Display scope warning

The safety pattern is propose, explain, preserve. The system may propose a correction or expansion but must show evidence, preserve the original and avoid committing uncertain transformations to the clinical record.

H.12 OCR and code-mix annotation guide

Annotation label	Example	Instruction
OCR substitution	5OOmg for 500mg	Mark observed string and candidate separately
OCR segmentation	nephrec- / tomy	Record line boundary and evidence for joining
Diacritic loss	Romanised variant	Retain surface; do not invent canonical form
Code-switch boundary	patient ku fever	Label token language/role where possible
Regional inflection	irundhuchu	Use qualified language reviewer for lemma/features
Mixed entity	paracetamol 500mg kuduthen	Link drug/strength span; retain surrounding tokens
Ambiguous transliteration	One spelling, multiple words	Keep alternatives and contextual judgement

At least two annotators fluent in the regional language should define transliteration variants and morphological segmentation. Character n-gram language identification provides a distribution, not truth, especially for short tokens. Evaluation should hold out writers and note templates to test generalization rather than memorization.